

INTERNET OF THINGS AND ARDUINO CLOUD

INTERNSHIP REPORT

Submitted by

RAVAL TILAK

211250107518

BACHELOR OF ENGINEERING

In

Computer Engineering

SHREE SWAMINARAYAN INSTITUTE OF TECHNOLOGY, BHAT



Gujarat Technological University, Ahmedabad

August, 2023



SHREE SWAMINARAYAN INSTITUTE OF TECHNOLOGY, BHAT

Gandhinagar - Ahmedabad Rd, Near EDI & Sardar Patel Ring Road Circle
, GIDC Bhat, Bhat, Gandhinagar, Gujarat 382428

CERTIFICATE

This is to certify that the internship report entitled “**INTERNET OF THINGS AND CLOUD**” has been carried out by **RAVAL TILAK** under my guidance in **Computer Engineering**, 7th Semester of Gujarat Technological University, Ahmedabad during the academic year 2023-24.

Prof. Khushboo Vyas

Internal Guide
Computer Engineering
SSIT, Bhat, Gandhinagar

Prof. Darshan Patel

Head of Department
Computer Engineering
SSIT, Bhat, Gandhinagar

GTU COMPLETION CERTIFICATE**GUJARAT TECHNOLOGICAL UNIVERSITY**

CERTIFICATE FOR COMPLETION OF ALL ACTIVITIES AT ONLINE PROJECT PORTAL

B.E. SEMESTER VII, ACADEMIC YEAR 2023-2024

Date of certificate generation : 25 October 2023 (23:12:21)

This is to certify that, **Raval Tilak** (Enrolment Number - 211250107518) working on project entitled with **IOT & Arduino Cloud** from **Computer Engineering** department of **SHREE SWAMINARAYAN INSTITUTE OF TECHNOLOGY, BHAT, GANDHINAGAR** had submitted following details at online project portal.

Internship Project Report	Completed
---------------------------	-----------

Name of Student : Raval Tilak

Name of Guide : Mrs. Khushboo Meet Vyas

Signature of Student : _____

*Signature of Guide : _____

Disclaimer :

This is a computer generated copy and does not indicate that your data has been evaluated. This is the receipt that GTU has received a copy of the data that you have uploaded and submitted as your project work.

*Guide has to sign the certificate, Only if all above activities has been Completed.

COMPANY OFFER LETTER



Office No.204, Second Floor,
Goldmine Jewel, B/h. D-Mart,
Sector-26, Gandhinagar,
Gujarat - INDIA

Ref.:

Date: 27-July-2023

To,
Prof. Darshan Patel
HOD, CE Department
Shree Swaminarayan Institute of Technology,
Bhat, Gandhinagar.

Sub: Offer letter for 15 days internship in IOT & Arduino Cloud to **Raval Tilak**

Ref: (1) Interview for 15 days internship in IOT & Arduino Cloud at Gandhinagar office. Dt 27-7-2023 to 10-8-2023.

Madam /Sir,

With reference to above subject & letter under reference, **Raval Tilak, BE CE**, is selected for 15 days internship training in Internet Of Things and Arduino Cloud for the course of **BE in Computer Engineering** in our organization as per below terms and conditions:

- (1) Details collected for the project should not be used for any other purpose by the student.
- (2) Permission is granted at their risk and cost and there should not arise any legal complications from their findings.
- (3) Student has to observe rules & regulations of the company during their visit for the above purpose.
- (4) Student has to submit their final project report to company without fail.

This is for your information,

Thanking You,



Yours Faithfully

Director
TECHInfinity
Gandhinagar

Copy To:
- **Raval Tilak** (Enrollment no:211250107518)

Phone: +91 90 81 81 45 00

Email: info@techinfinity.org

www.techinfinity.org

COMPANY CERTIFICATE



Office No.204, Second Floor,
Goldmine Jewel, B/h. D-Mart,
Sector-26, Gandhinagar,
Gujarat - INDIA

Date: 11-August-2023

TO WHOMSOEVER IT MAY CONCERN

Sub: Certificate of Completion of 15 days internship in IOT & Arduino Cloud by **Raval Tilak**
Ref: 15 days internship in IOT & Arduino Cloud at Gandhinagar office. Dated 27-7-2023 to 10-8-2023.

With reference to above subject & letter under reference, **Raval Tilak, BE CE**, has successfully completed his 15days internship training in Internet Of Things and Arduino Cloud for the course of **BE in Computer Engineering** in our organization.

During the internship he demonstrated good design and development skills to complete the project with a self-motivated attitude to learn new things. His performance exceeded expectations and was able to complete the project successfully on time.

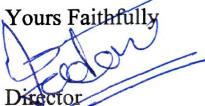
We wish **Mr. Raval Tilak** the very best for his future endeavours.

This is for your information,

Thanking You,



Yours Faithfully


Director
TECHInfinity
Gandhinagar

Phone: +91 90 81 81 45 00

Email: info@techinfinity.org

www.techinfinity.org

ACKNOWLEDGEMENT

The successful completion of my summer internship named “**IOT & Cloud**” stands on the constant encouragement, guidance, and support, both technical & personal of many individuals.

However, it would not have been possible without the kind support and help of many individuals and institutional. We would like to extend our sincere thanks to all of them. We are highly indebted to our internal guide **Prof. Khushboo Vyas** for their guidance and constant supervision as well as for providing necessary support in the summer internship. This project work could not have been completed without the expert guidance of **Prof. Khushboo Vyas** who shown me proper path and direction of work.

They opened for me the offer of knowledge from repository of their experience. They have been encouraging me throughout my project work which increased my confidence and gave me the strength of working hard to achieve my target in time. This has impelled me to express my sincere gratitude to my project coordinator **Mrs. Sejal Jadav** Ma'am.

At the same time, I would like to thank my Head of the Department **Prof. Darshan Patel** for his kind cooperation and support. I would also appreciate his keen interest in helping me and his regular guidance throughout the project. I would also like to thank all the staff members of my department for their cooperation and help me when asked for. I am also grateful to all my colleagues for helping me with all their resources and knowledge they could, as when required.

RAVAL TILAK

(21125010751

ABSTRACT

The main purpose of the project is to Make/Learning a IOT & Cloud based project, by using that we can manage and operate IoT Devices with the help of cloud. This project introduces a novel approach to home automation by combining the power of the Internet of Things (IoT) and cloud computing with Amazon's Alexa voice assistant. The aim is to create an integrated ecosystem where everyday household devices are seamlessly interconnected, accessible remotely, and easily controllable through voice commands.

Using a variety of IoT devices such as smart switches, sensors and appliances, a network is established within the home environment. These devices communicate with a central hub, utilizing Wi-Fi, Protocols, which then interfaces with a cloud-based platform. This cloud infrastructure serves as a hub for managing and controlling the IoT devices, offering real-time monitoring and user-friendly remote access via a dedicated mobile Arduino Cloud Dashboard.

In summary, the "IoT and Cloud-Based Home Automation Using Alexa" project amalgamates cutting-edge technologies to create an intelligent and user-centric home automation solution. By unifying IoT, cloud computing, and voice control, the project demonstrates the potential of these technologies to redefine how we interact with and manage our living spaces, ultimately enhancing comfort efficiency.

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1. INTRODUCTION

For newly introduced subject “summer internship (3170001)” by GTU. I joined one well known Gandhinagar based IT/IOT company named “**TECHInfinity**” to learn “**IOT & Arduino Cloud**” from scratch. The duration of the internship was about 15 days. In this time interval, I learned a technical thing regarding IOT and Arduino Cloud with tools and technologies such as Microcontroller, Hardware Knowledge (Soldering, PCB, Embedded Systems) etc.

With abstract of project is **Home Automation using Arduino Cloud**.

In the point of fact,

I got the benefit of having brilliant external guide as well as internal guide from my collage. I explored all the aspects of Internet of Things Components & framework in detail as well as I got the chance my existing knowledge for brush up the skills in Arduino Cloud, Soldering, Basic PCB Design.

1.1) Project summary

The "Home Automation using Arduino Cloud and Alexa" project is a groundbreaking endeavor that merges the power of IoT and voice recognition technology to create a smart and responsive living environment. This project aims to enhance the convenience, energy efficiency, and security of homes by seamlessly integrating smart devices through Arduino Cloud's IoT platform and controlling them using Amazon's Alexa voice assistant.

Objective:

The primary goal of this project is to design and implement a comprehensive home automation system that utilizes Arduino Cloud for device connectivity and data management, while leveraging Alexa's voice control capabilities for effortless user interaction.

Key Features:

- Device Integration:
- IoT Connectivity:
- Voice Control:
- Customization:
- Energy Efficiency:
- User Convenience:
- Energy Savings:
- Future Scope:

This project lays the foundation for advanced home automation innovations, including more sophisticated device interactions, predictive analytics for energy optimization, and integration with other smart home ecosystems.

2. ABOUT TECHNOLOGY

2.1 IOT

The Internet of Things (IoT) refers to the network of interconnected devices, objects, and systems that communicate and exchange data with each other over the internet. This technology has gained immense traction over the past few years, transforming various industries and aspects of our daily lives. At its core, IoT involves embedding sensors, software, and other technologies into everyday objects to enable them to gather and transmit data, as well as interact with their environment or other devices. This interconnectedness allows for enhanced automation, data analysis, and decision-making.

2.2 Arduino Cloud

Arduino Cloud is a platform provided by Arduino, a well-known open-source electronics platform that allows individuals and developers to create interactive electronic projects. Arduino Cloud serves as a vital component for building and managing Internet of Things (IoT) projects, offering a range of tools and services that simplify the process of connecting and controlling IoT devices.

2.3 Soldering & PCB

Soldering: Soldering is a technique that fuses metal components using a heated soldering iron, creating strong electrical connections in electronics assembly.

PCB: (Printed Circuit Board): PCB's are essential in electronics, providing a platform for components to be interconnected, facilitating the creation of intricate circuits that power modern devices.

2.4 Voice Assistant (Alexa)

Alexa: Alexa, developed by Amazon, is an intelligent virtual assistant that uses voice recognition to perform tasks, answer questions, and control smart devices, enhancing the convenience of daily life.

2.5 Why Use Arduino Cloud?

Arduino Cloud: is a platform provided by Arduino, a widely used open-source electronics platform, that offers tools and services for managing and controlling Internet of Things (IoT) devices, including those used in home automation systems. Home automation involves the integration of various devices and technologies to create a smart home environment that enhances convenience, energy efficiency, and overall living experience. Here's why Arduino Cloud might be used for home automation:

2.6 Features of Arduino Cloud.

Device Management: Arduino Cloud provides a centralized dashboard for managing all your connected IoT devices. This includes configuring, monitoring, and updating these devices remotely. For a home automation setup, this means you can control your smart lights, thermostats, security cameras, and other devices from a single interface.

Ease of Integration: Arduino Cloud offers a user-friendly interface and supports a wide range of Arduino boards and IoT hardware, making it easier to connect and configure your devices. This is especially valuable for people with limited programming or electronics experience.

Remote Control: With Arduino Cloud, you can remotely control your connected devices from anywhere with an internet connection. This enables you to adjust your home's settings, such as lighting, temperature, or security, even when you're not physically present.

Automation and Triggers: You can set up automation routines and triggers based on specific conditions or events. For example, you could automate your lights to turn on when motion is detected or to adjust the thermostat based on the time of day.

Data Monitoring and Insights: Arduino Cloud allows you to collect and analyze data from your IoT devices. This can help you gain insights into energy consumption patterns, occupancy trends, and other relevant information, enabling you to make informed decisions to optimize your home's efficiency.

Scalability: Whether you have a few devices or a complex network of interconnected devices in your home automation system, Arduino Cloud can accommodate your needs. It's designed to scale as your requirements grow.

Customization: While Arduino Cloud offers user-friendly tools for those who prefer a simpler setup, it also provides options for more advanced users to customize their automation and control logic using programming languages like C++. In summary, Arduino Cloud is a convenient and versatile platform for managing and controlling IoT devices in a home automation context. It simplifies the process of connecting, configuring, and interacting with your smart devices, making it easier to create a seamless and efficient smart home environment

3. SYSTEM REQUIREMENT AND SPECIFICATIONS

3.1 Software Requirements:

Operating System:	Windows, Linux Or MacOS
Software:	Arduino IDE
Web Server:	Browser (To Access Cloud).

3.2 Hardware Requirements:

Microcontroller	Arduino Development Board Or ESP8266/32
Sensors	Basic Knowledge of Sensors. And Sensors Based On project.

4. TASK/PROJECT SCREENSHOT WITH CODE

4.1 TASK-1

What Is Arduino IOT Cloud ? & How to Setup To Cloud ?

Arduino IoT Cloud is a platform provided by Arduino that enables users to easily develop and manage Internet of Things (IoT) projects. It offers tools and services for connecting, controlling, and monitoring IoT devices through the internet. With Arduino IoT Cloud, developers can create applications that involve remote control, data visualization, and automation of devices, all while abstracting much of the complexity of IoT development. This platform provides an environment for connecting Arduino boards and other compatible devices to the cloud, making it accessible for both beginners and experienced developers to build IoT solutions.

Setting UP Arduino Cloud:

Introduction:

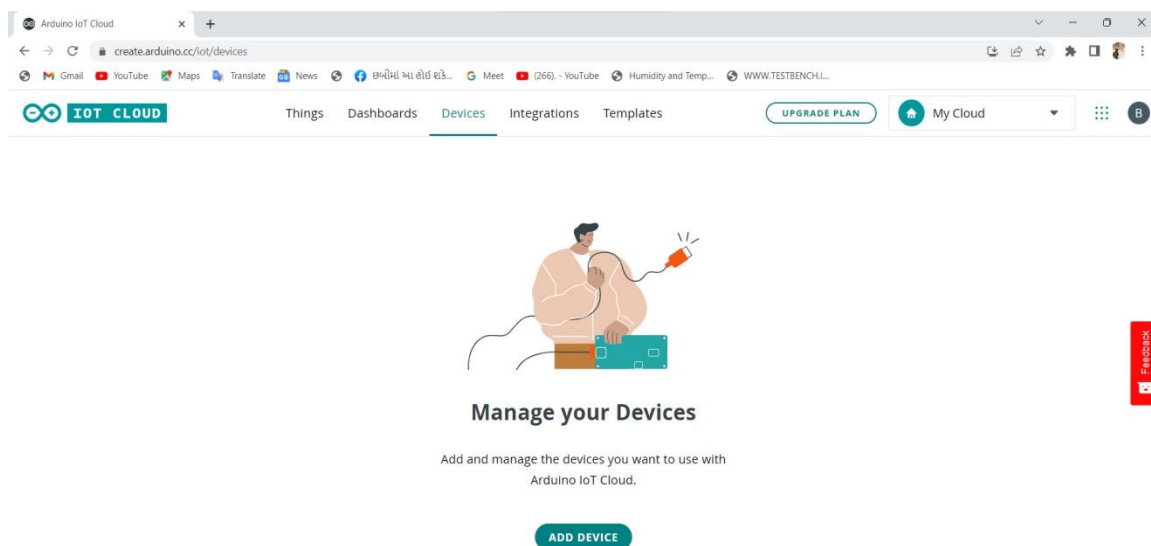
Connected devices around the world are increasing by billions every year. The Arduino IoT Cloud is a platform that allows anyone to create IoT projects, with a user friendly interface, and an all in one solution for **configuration, writing code, uploading** and **visualization**.

Here in this documentation, we will go through the steps necessary to set up an ESP8266 board with the Arduino Cloud IoT.

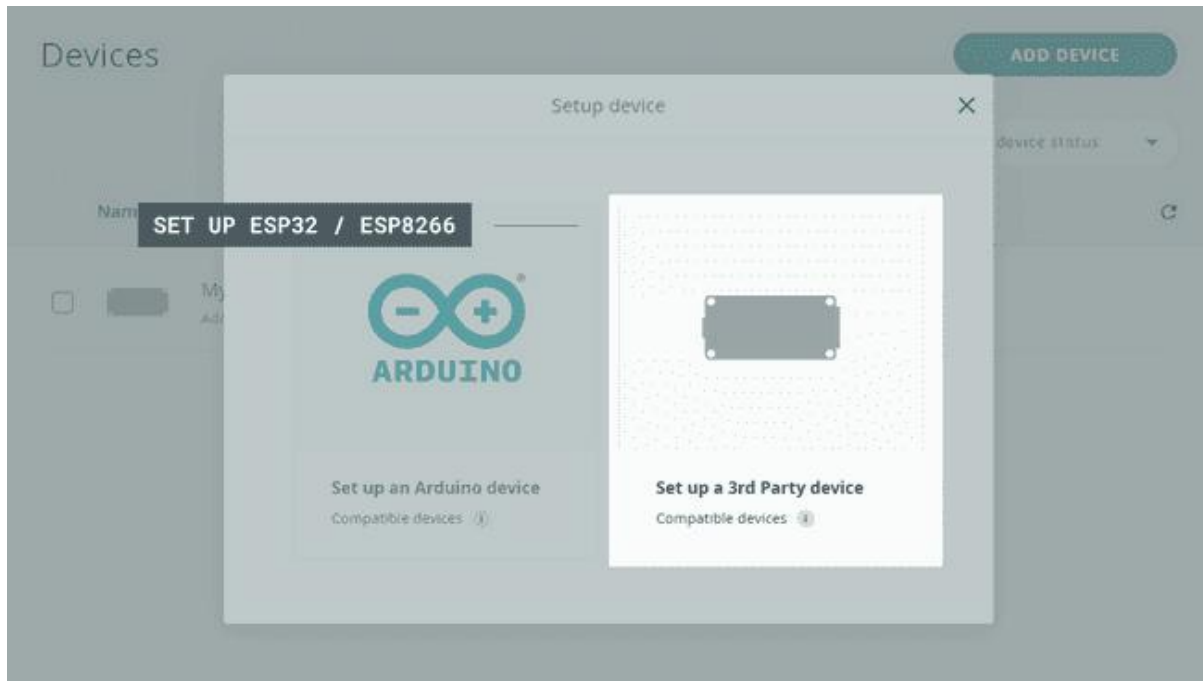
Here are some steps to start with Arduino cloud IoT:

Step 1: Setting up the device

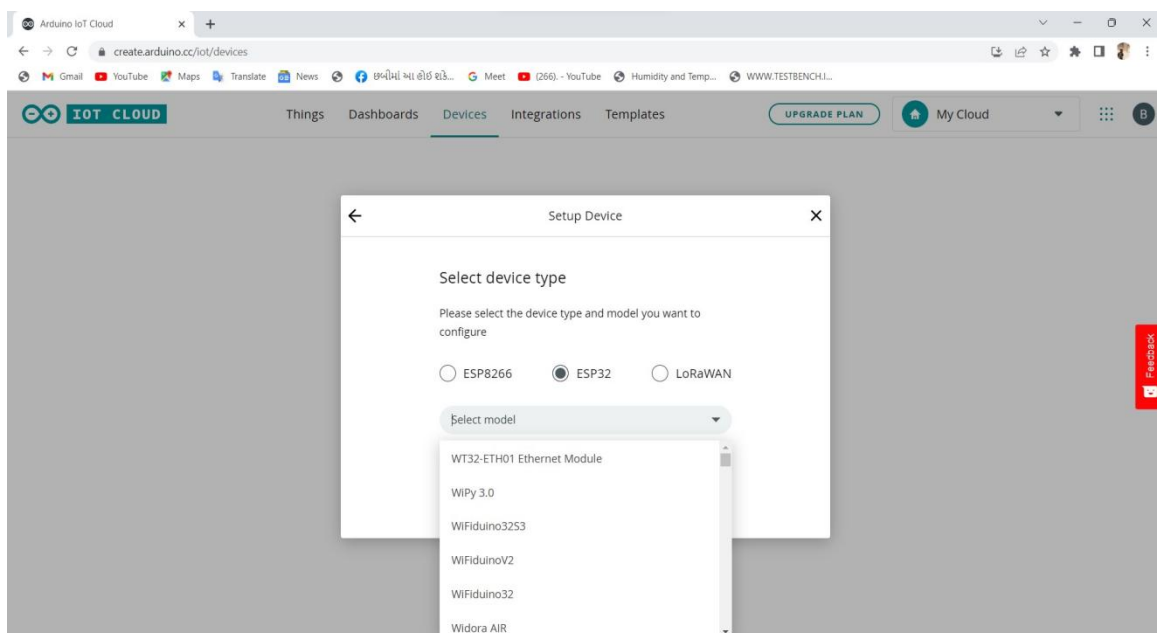
i) Once in the cloud, click on the **"Devices"** tab. Then, click on the **"Add device"** button.



ii) Then, click on **"Set up a 3rd party device"**.

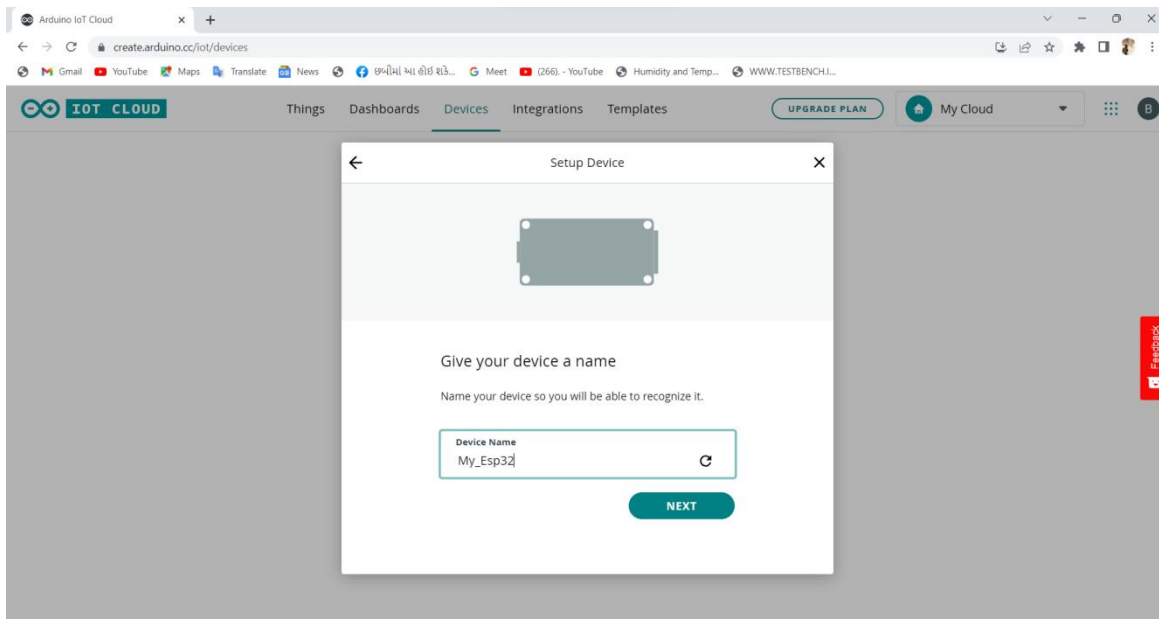


iii) Now we need to select what board we are using. Select either ESP32 or ESP8266, and then choose the board from the drop-down menu.

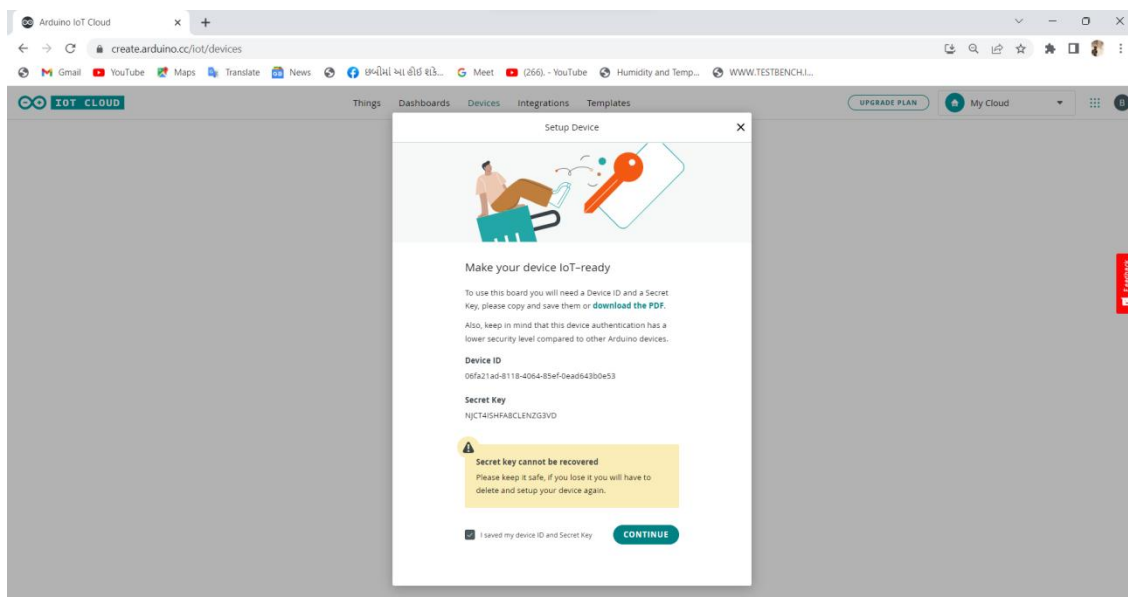


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iv) Now the board needs a name.

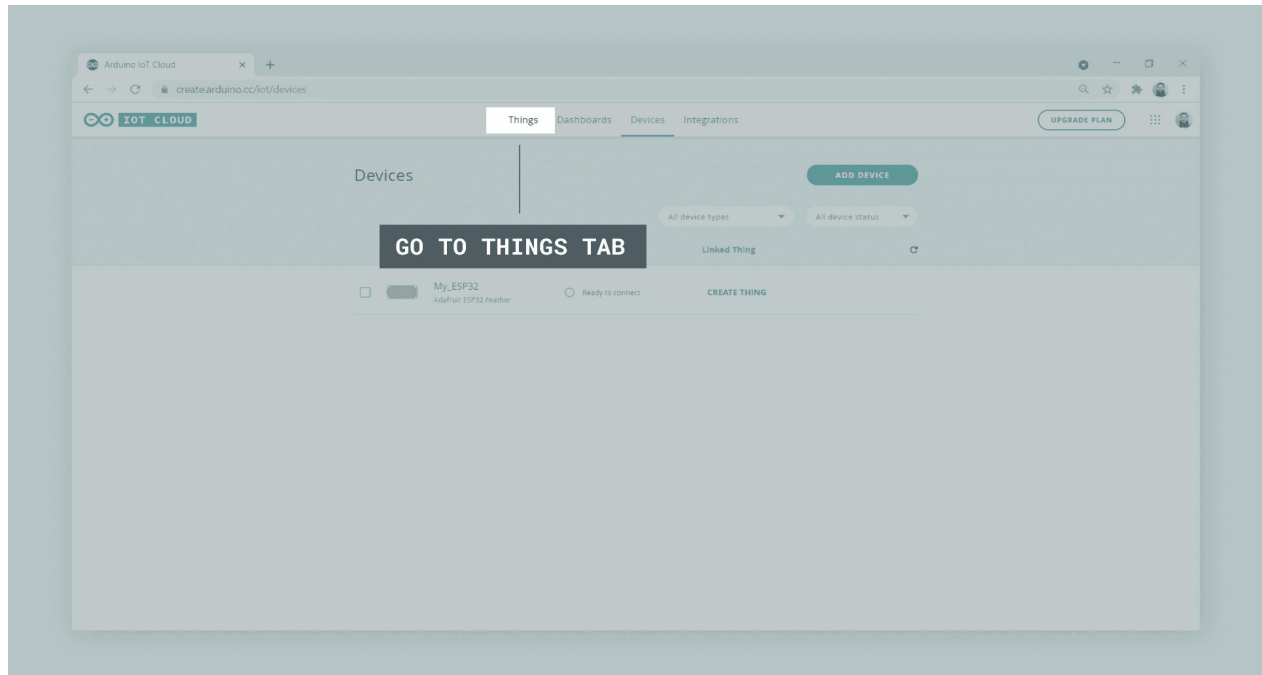


v) You will now receive your **Device ID** and **Secret key**. Please note that the secret key cannot be recovered, so make sure you note it down. You can also download a PDF with the information. When you have saved it, tick the box at the bottom, and click on "**Continue**".

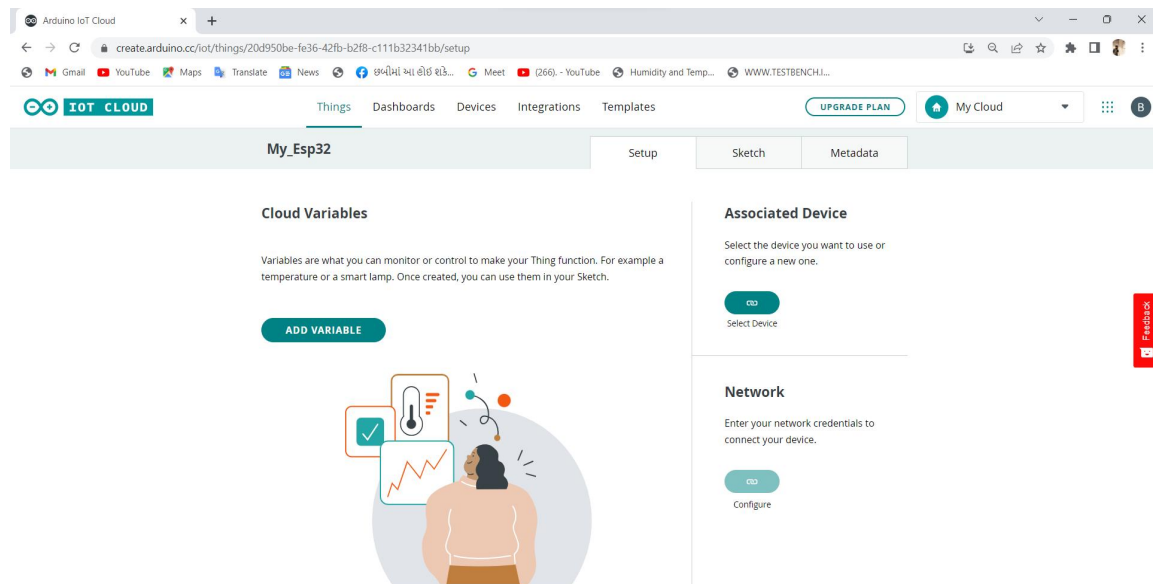


Step 2: Creating a Thing

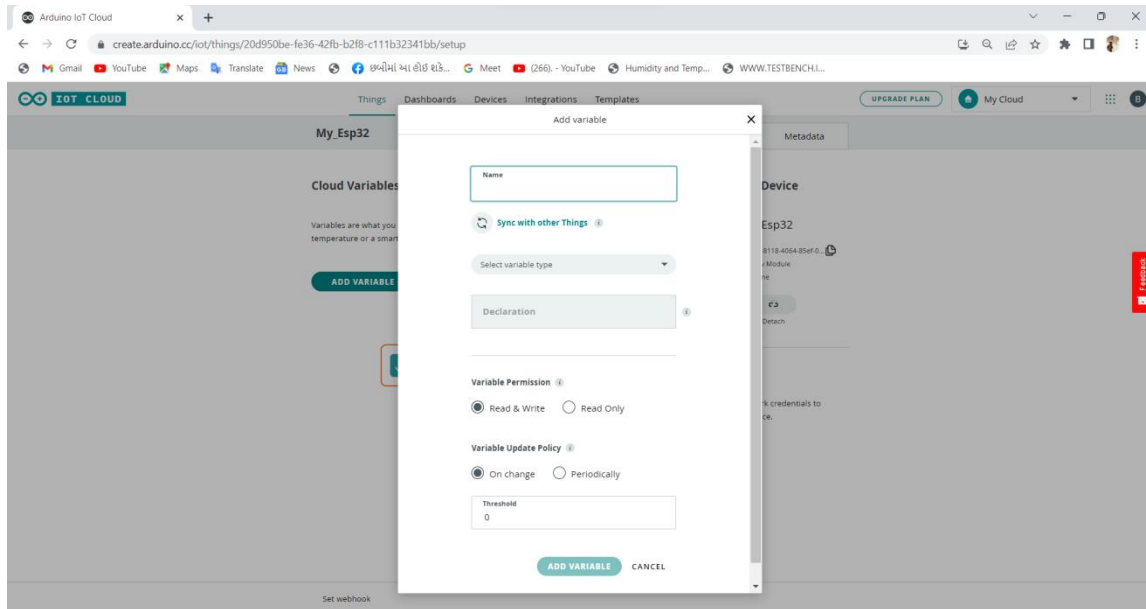
The next step is to create a Thing. This is done by navigating to the **"Things"** tab.



By clicking on “Things” this type of window will appear.



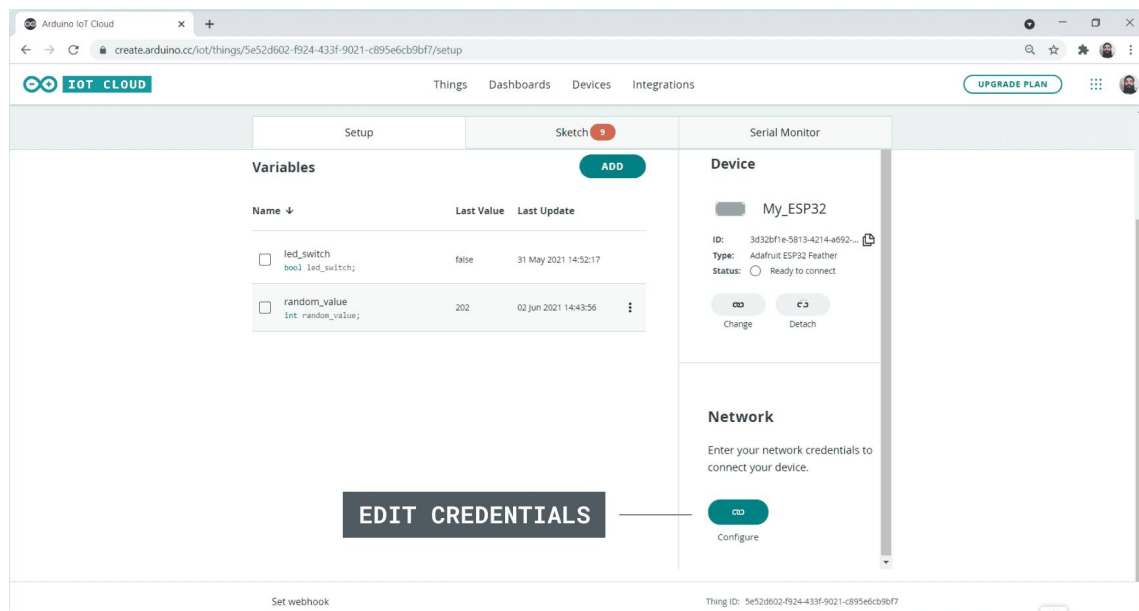
Now, we need to link our device with our Thing. This is done by clicking on the button in the **"Device"** section. This will open a window, where your ESP32 should be available to select. Now, after associate the device successfully, click on the **“ADD VARIABLE”** to **create variables** for our project.

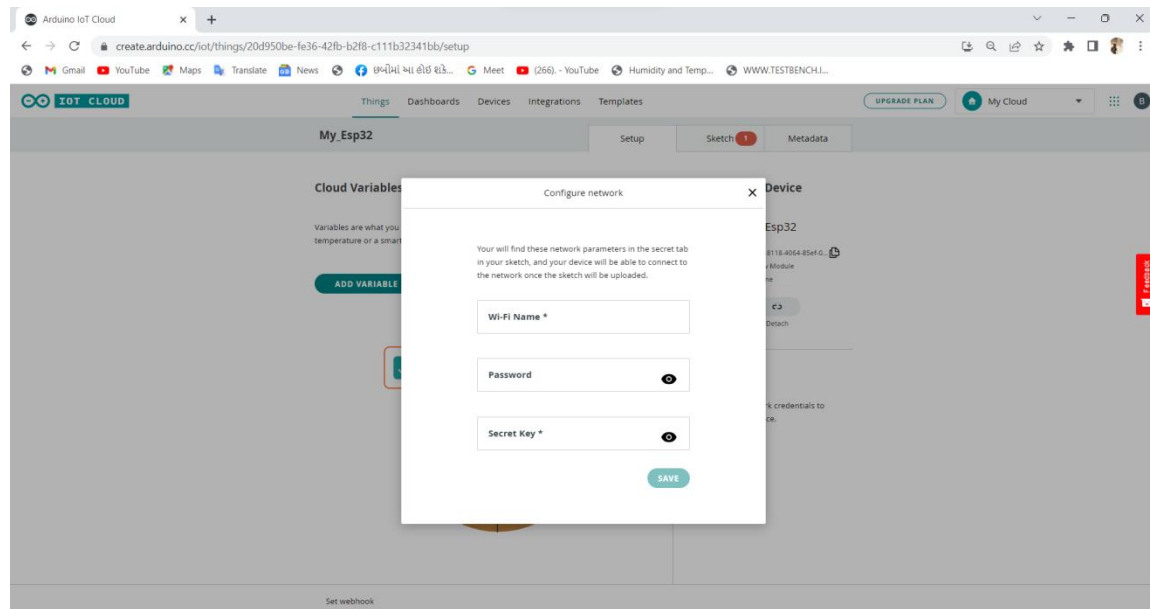


Step 3: Adding credentials

Now, we need to enter the credentials for our network, plus the secret key generated during the device configuration.

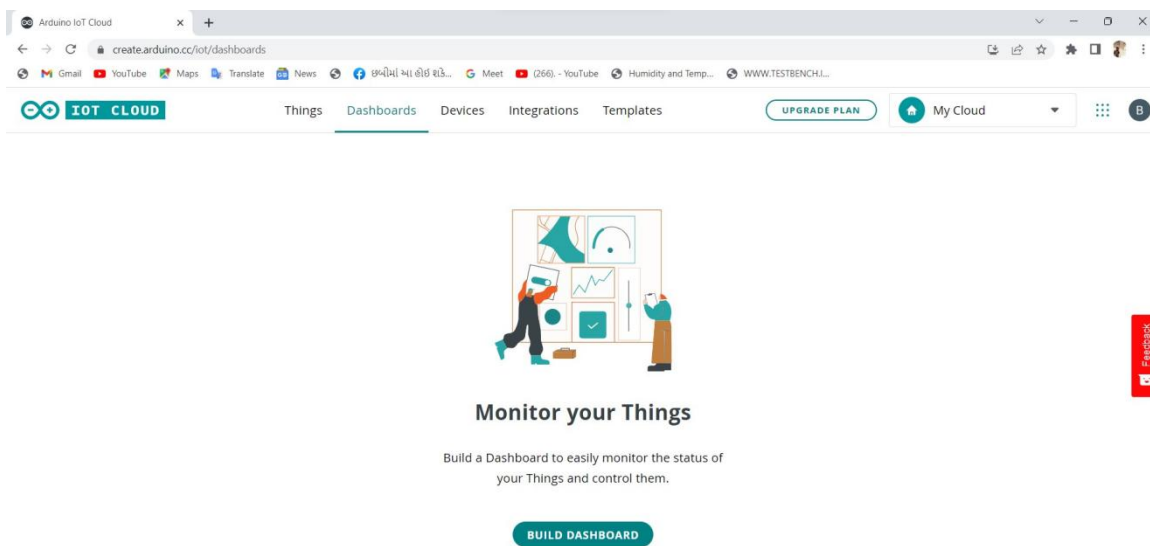
First, click on the button in the "Network Section".



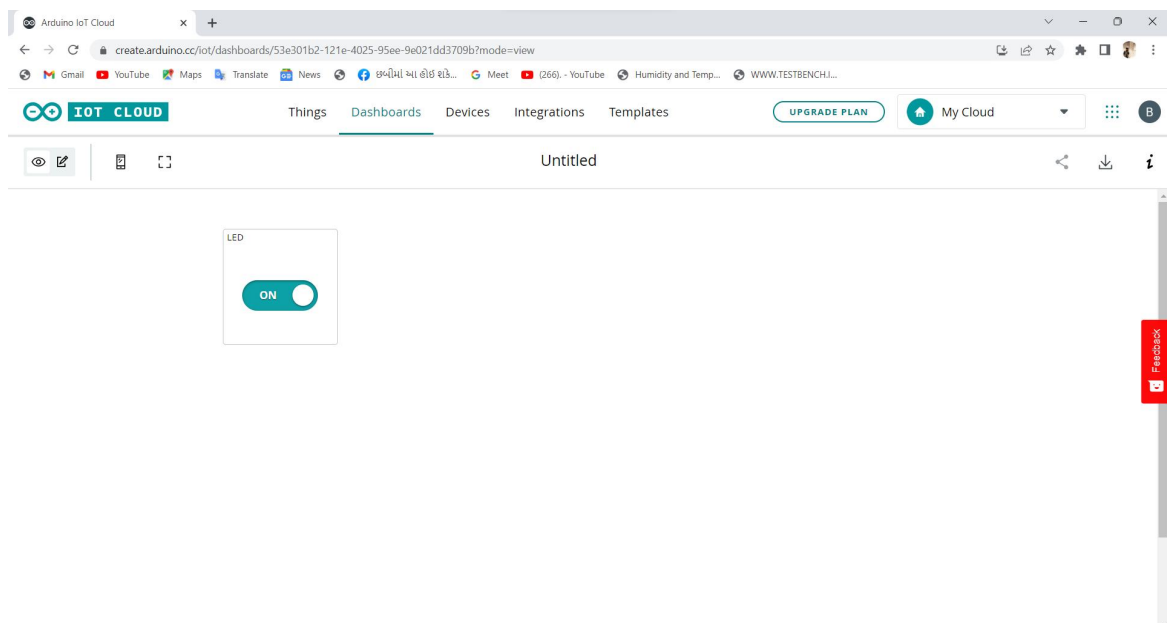
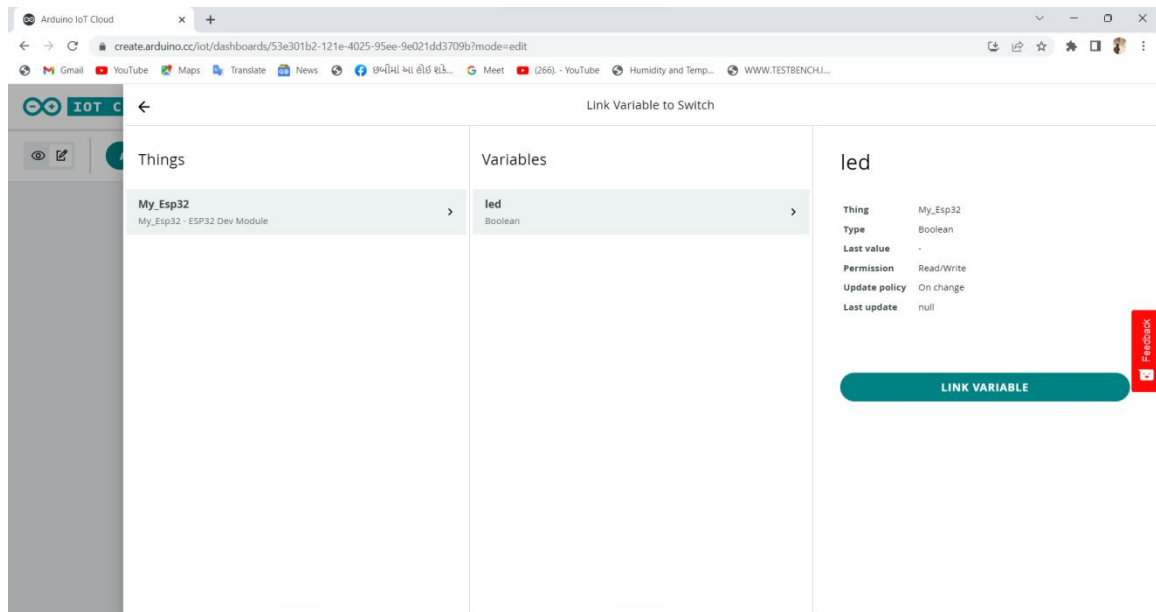


Step 4: Creating dashboard

First, click on the “Dashboard” to create dashboard for the project.



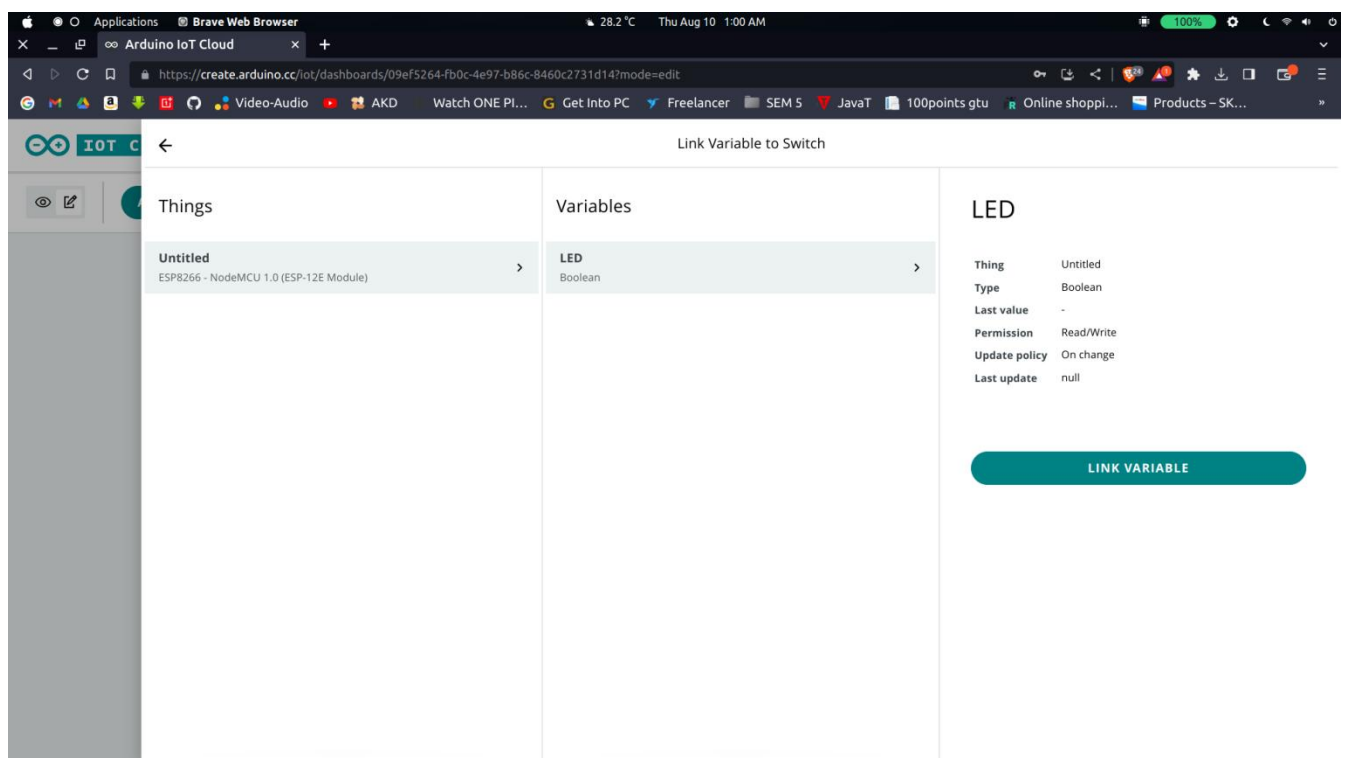
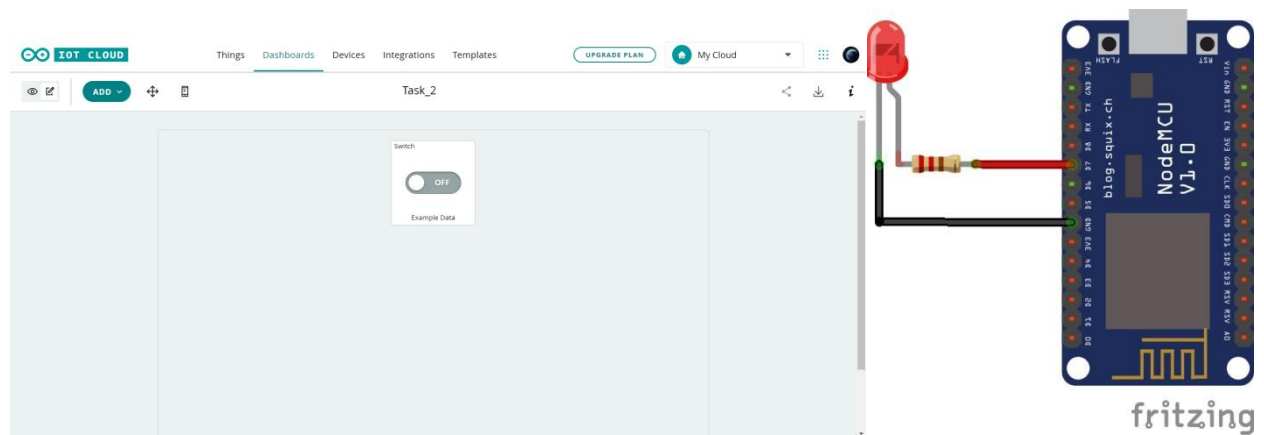
Now, click on the “ADD” to add widgets for our project and link the variables according to the widgets.

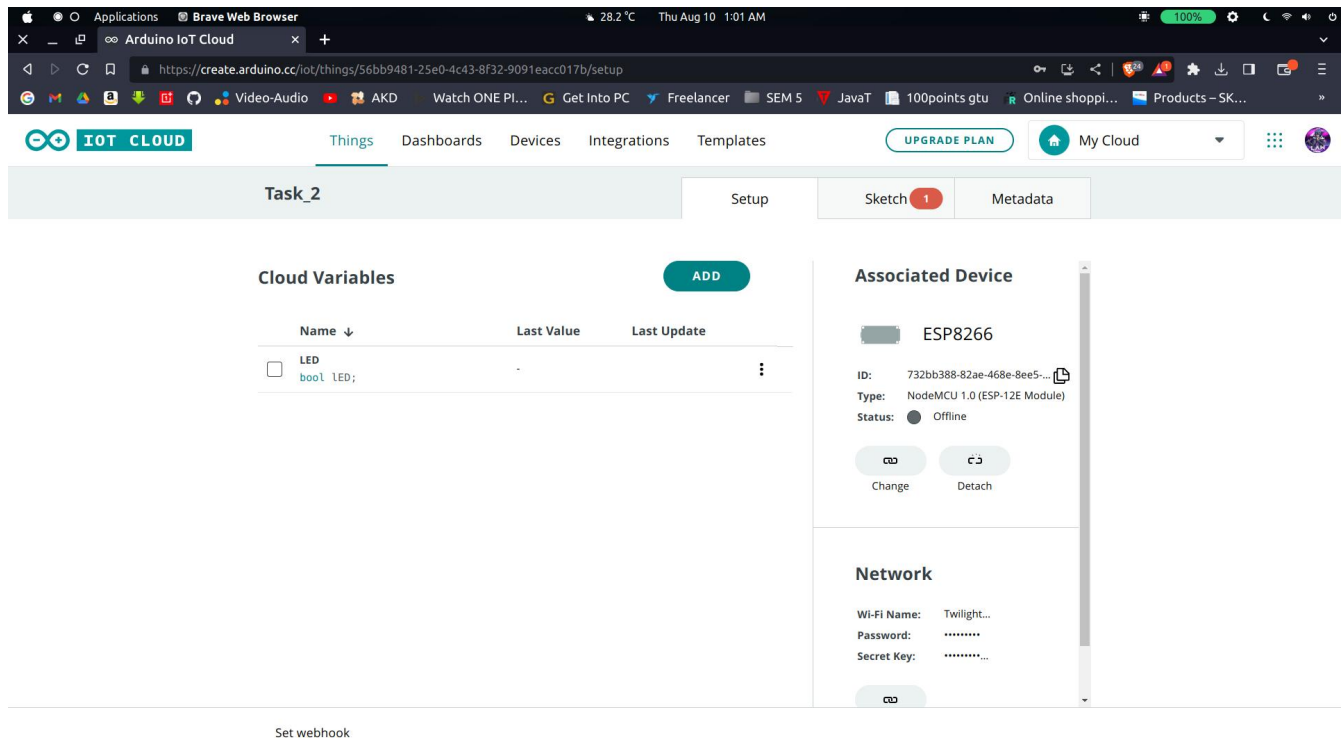


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4.2 Task -2

Blink LED Using Arduino Cloud :





Code:-

```
#include "arduino_secrets.h"
#include "thingProperties.h"
int LED = 0;
```

```
void setup() {
  Serial.begin(9600);
  delay(1500);
  initProperties();
  pinMode(LED, OUTPUT);
```

```
  ArduinoCloud.begin(ArduinoIoTPreferredConnection)
;
  setDebugMessageLevel(2);
  ArduinoCloud.printDebugInfo();
}
```

```
void loop() {
  ArduinoCloud.update();
}
```

```
void onLEDChange() {
  digitalWrite(LED, LED);
}
```

```
-----#include"thingProperties.h"-----
#include <ArduinoIoTCloud.h>
#include <Arduino_ConnectionHandler.h>

const char DEVICE_LOGIN_NAME[] = "732bb388-82ac-468e-8ee5-af94cb49f150";
```

```
const char SSID[]          = SECRET_SSID;
const char PASS[]          =
SECRET_OPTIONAL_PASS;
const char DEVICE_KEY[] =
SECRET_DEVICE_KEY;
```

```
void onLEDChange();
```

```
bool LED;
```

```
void initProperties(){
```

```
  ArduinoCloud.setBoardId(DEVICE_LOGIN_NAME);
  ArduinoCloud.setSecretDeviceKey(DEVICE_KEY);
  ArduinoCloud.addProperty(LED, READWRITE,
ON_CHANGE, onLEDChange);
```

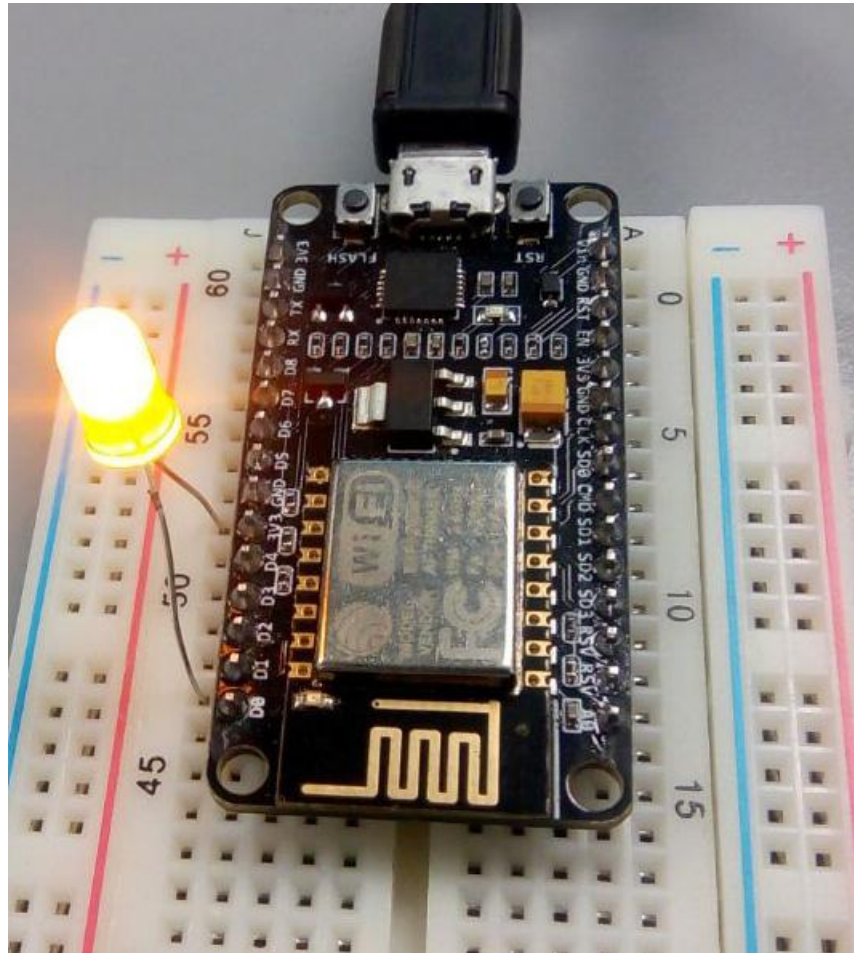
```
}
```

```
WiFiConnectionHandler
ArduinoIoTPreferredConnection(SSID, PASS);
```

```
-----#include"arduino_Secrets.h"-----
```

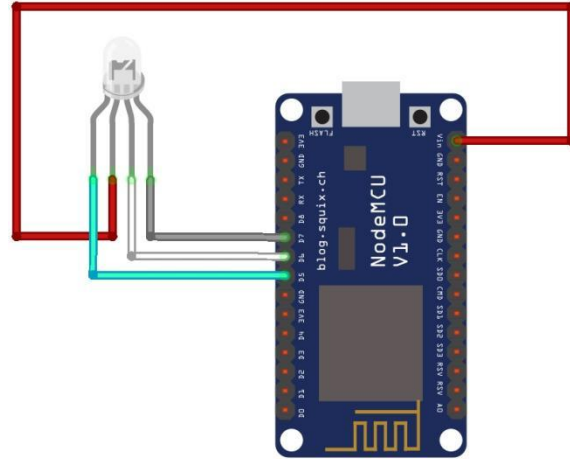
```
#define SECRET_SSID "" // Your Wifi Credentials  
#define SECRET_OPTIONAL_PASS ""  
#define SECRET_DEVICE_KEY ""
```

Output : -

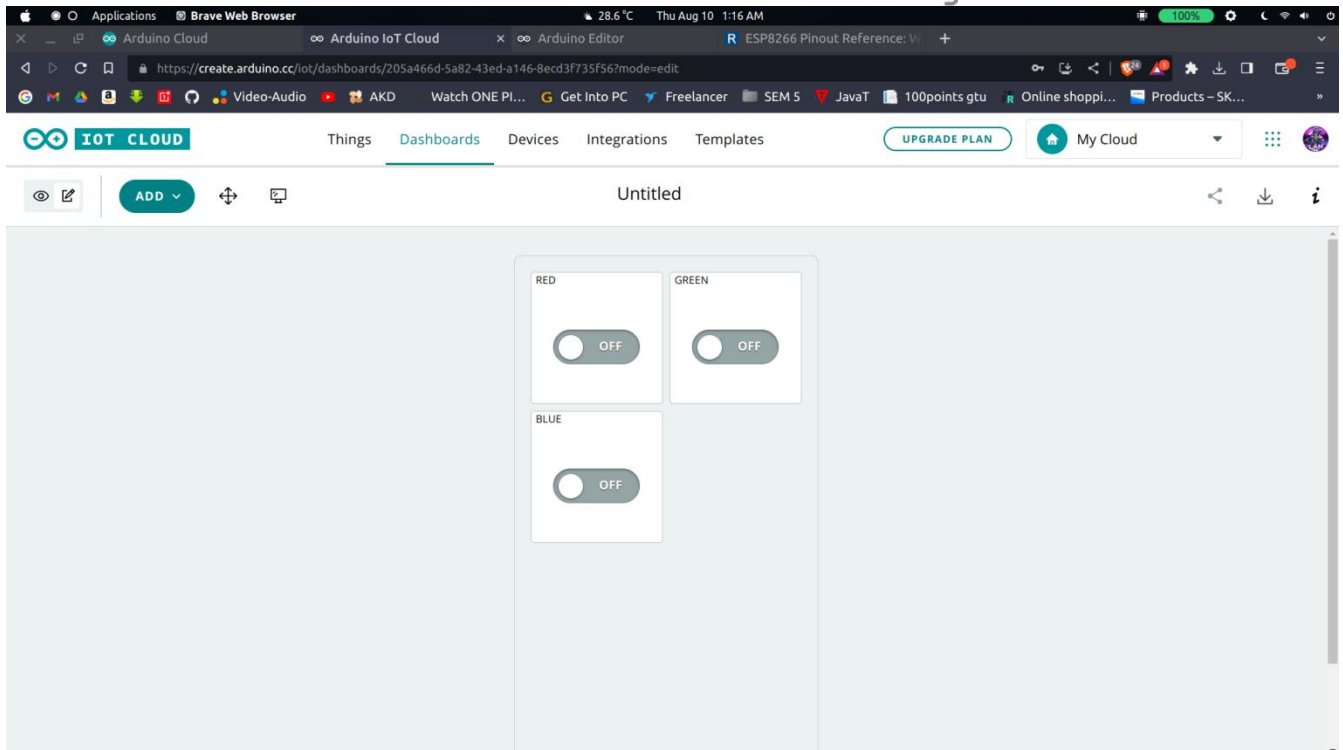


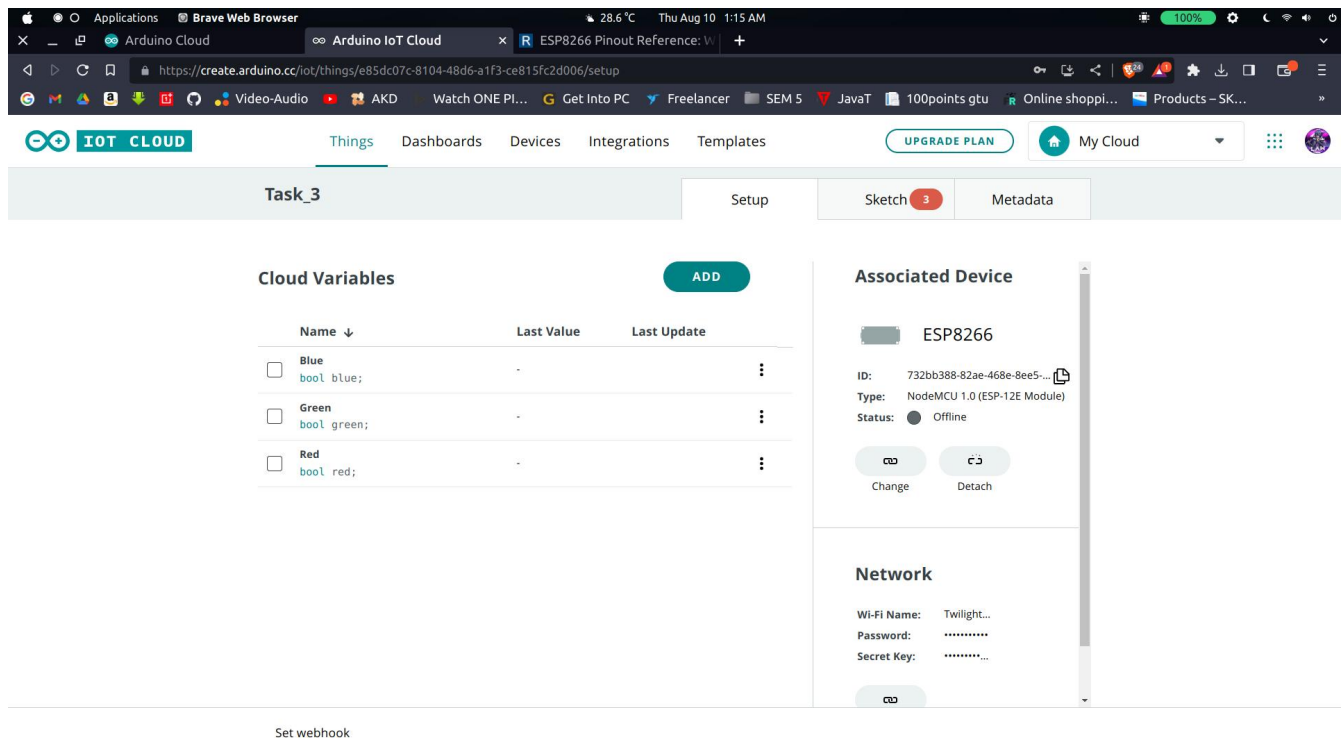
4.3 TASK 3: -

RGB LED Using Arduino Cloud.



fritzing





Code :-

```
#include "arduino_secrets.h"
#include "thingProperties.h"
```

```
bool Red;
bool Green;
bool Blue;
```

```
int R_LED = D1;
int G_LED = D2;
int B_LED = D3;
```

```
void setup() {
  Serial.begin(9600);
  delay(1500);
  initProperties();
```

```
  pinMode(Red, OUTPUT);
  pinMode(Green, OUTPUT);
  pinMode(Blue, OUTPUT);
  pinMode(R_LED, OUTPUT);
  pinMode(G_LED, OUTPUT);
  pinMode(B_LED, OUTPUT);
```

```
  ArduinoCloud.begin(ArduinoIoTPreferredConnection)
;
  setDebugMessageLevel(2);
```

```
  ArduinoCloud.printDebugInfo();
}
```

```
void loop() {
  ArduinoCloud.update();
}

void onRedChange() {
  digitalWrite(R_LED, red);
}

void onBlueChange() {
  digitalWrite(G_LED, green);
}

void onGreenChange() {
  digitalWrite(B_LED, blue);
}
```

```
-----#include "arduino_Secrets.h"-----
----
```

```
#define SECRET_SSID ""
#define SECRET_OPTIONAL_PASS ""
#define SECRET_DEVICE_KEY ""
```

```

-----#include"thingProperties.h"-----

#include <ArduinoIoTCloud.h>
#include <Arduino_ConnectionHandler.h>

const char DEVICE_LOGIN_NAME[] =
"732bb388-82ae-468e-8ee5-af94cb49f150";

const char SSID[]          = SECRET_SSID;
const char PASS[]          =
SECRET_OPTIONAL_PASS;
const char DEVICE_KEY[] =
SECRET_DEVICE_KEY;

void onBlueChange();
void onGreenChange();
void onRedChange();

bool blue;
bool green;
bool red;

void initProperties(){

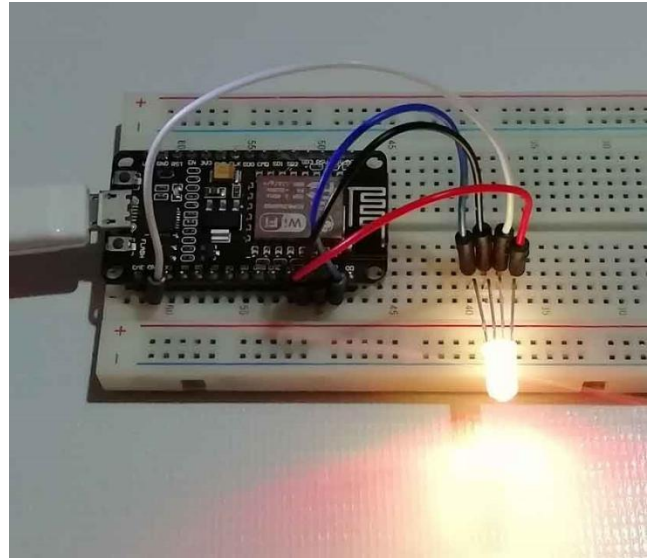
  ArduinoCloud.setBoardId(DEVICE_LOGIN_NAME)
;
  ArduinoCloud.setSecretDeviceKey(DEVICE_KEY);
  ArduinoCloud.addProperty(blue, READWRITE,
ON_CHANGE, onBlueChange);
  ArduinoCloud.addProperty(green, READWRITE,
ON_CHANGE, onGreenChange);
  ArduinoCloud.addProperty(red, READWRITE,
ON_CHANGE, onRedChange);

}

WiFiConnectionHandler
ArduinoIoTPreferredConnection(SSID, PASS);

```

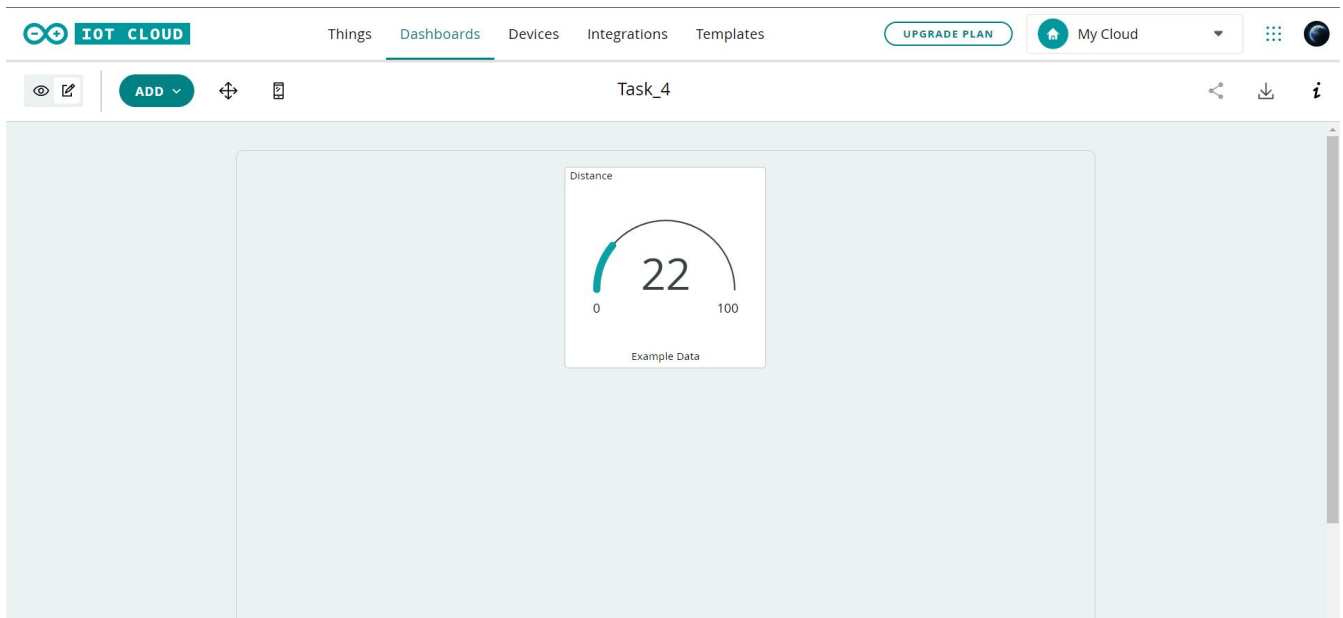
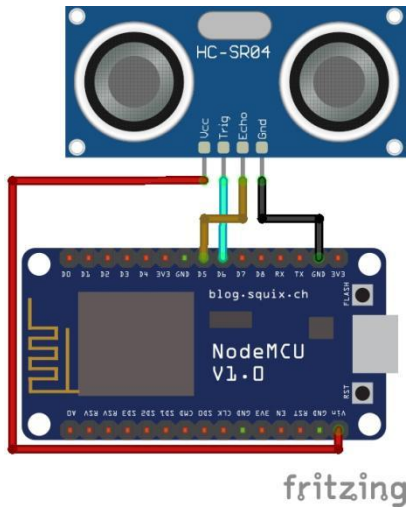
OUTPUT:-



3170001

4.4 TASK 4: -

Show Distance On Cloud Using Ultrasonic Sensor:



Code:

```

#include "arduino_secrets.h"
#include "thingProperties.h"

const int trigPin = D6;
const int echoPin = D5;

//define sound velocity in cm/uS
#define SOUND_VELOCITY 0.034

long duration;
float distance;

void setup() {
  Serial.begin(9600); // Starts the serial
  communication
  initProperties();
  pinMode(trigPin, OUTPUT); // Sets the trigPin as an
  Output
  pinMode(echoPin, INPUT); // Sets the echoPin as an
  Input

  ArduinoCloud.begin(ArduinoIoTPreferredConnection)
  ;
  setDebugMessageLevel(2);
  ArduinoCloud.printDebugInfo();
}

void loop() {
  ArduinoCloud.update();
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Reads the echoPin, returns the sound wave travel
  time in microseconds
  duration = pulseIn(echoPin, HIGH);

  // Calculate the distance
  distance = duration * SOUND_VELOCITY/2;

  // Prints the distance on the Cloud Dashboard
  Dist = distance;

  delay(1000);
}

void onDistChange() {
  // Add your code here to act upon Sensor1 change
}

```

```

-----#include"arduino_Secrets.h"-----

#define SECRET_SSID ""
#define SECRET_OPTIONAL_PASS ""
#define SECRET_DEVICE_KEY ""

-----#include"thingProperties.h"-----

#include <ArduinoIoTCloud.h>
#include <Arduino_ConnectionHandler.h>

const char DEVICE_LOGIN_NAME[] =
"732bb388-82ae-468e-8ee5-af94cb49f150";

const char SSID[]          = SECRET_SSID;
const char PASS[]          =
SECRET_OPTIONAL_PASS;
const char DEVICE_KEY[]    =
SECRET_DEVICE_KEY;

void onDistChange();

float Dist;

void initProperties(){

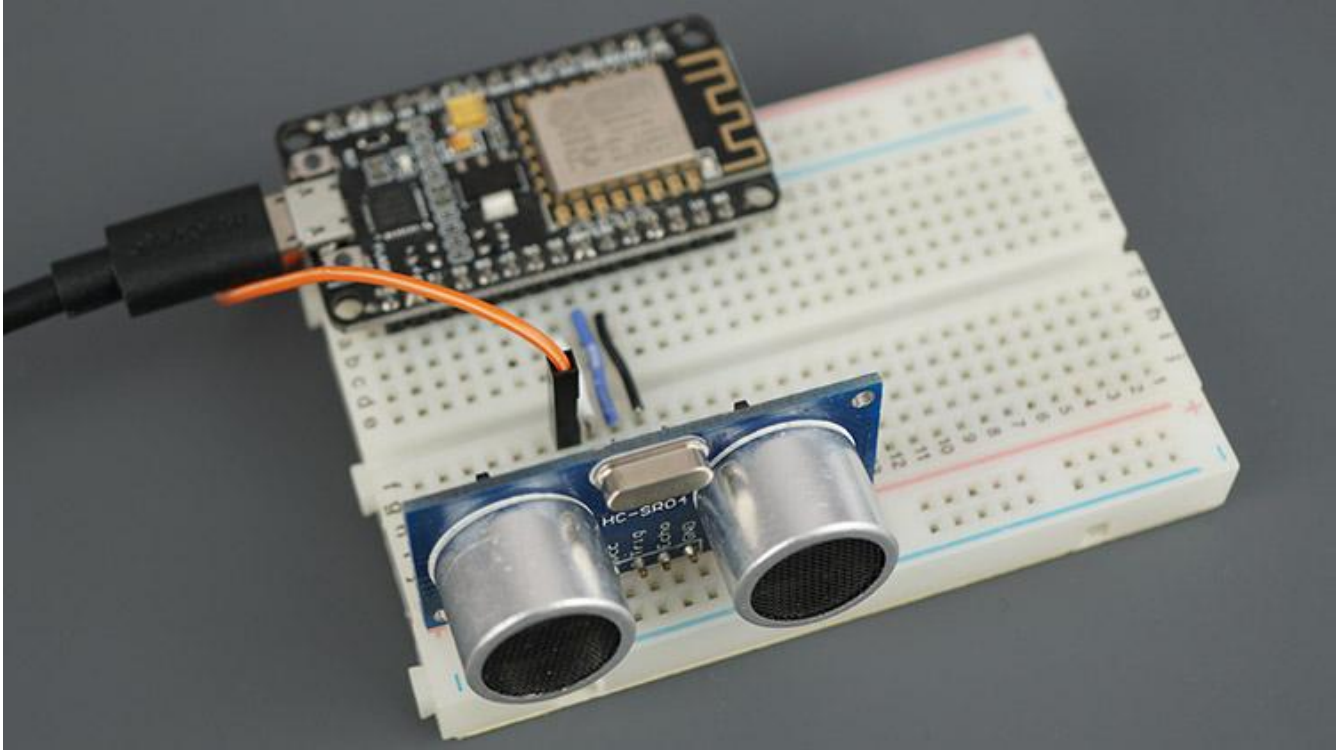
  ArduinoCloud.setBoardId(DEVICE_LOGIN_NAME)
  ;
  ArduinoCloud.setSecretDeviceKey(DEVICE_KEY);
  ArduinoCloud.addProperty(Dist, READWRITE,
  ON_CHANGE, onDistChange);

}

WiFiConnectionHandler
ArduinoIoTPreferredConnection(SSID, PASS);

```

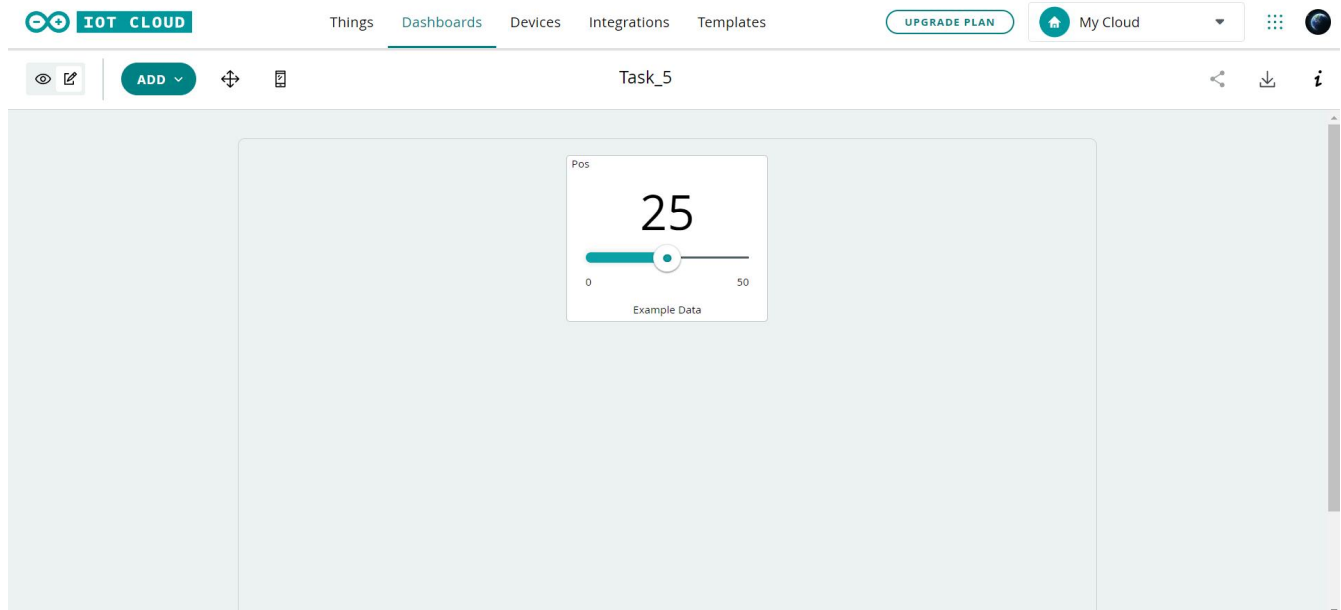
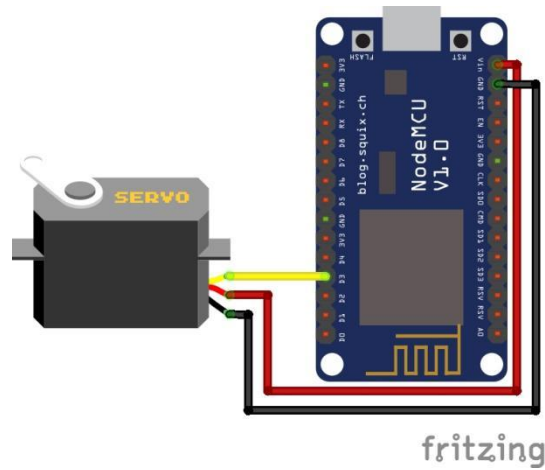
Output: -



3170001

4.5 Task 5: -

Control Servo Using Arduino IOT Cloud:



Code :

```

#include "arduino_secrets.h"

#include "thingProperties.h"

#include <Servo.h>

Servo myservo;

const int trigPin = D6;

const int echoPin = D5;

#define SOUND_VELOCITY 0.034

long duration;

float distance;

void setup() {

    Serial.begin(9600);    // Starts the serial
    communication

    myservo.attach(D3);

    initProperties();

    ArduinoCloud.begin(ArduinoIoTPreferredConne
    ction);

    setDebugMessageLevel(2);

    ArduinoCloud.printDebugInfo();

}

void loop() {

    ArduinoCloud.update();

    delay(100);

}

void onPositionChange() {

    // Add your code here to act upon Position change

    myservo.write(position);

}

```

```

-----#include"arduino_Secrets.h"-----

#define SECRET_SSID ""

#define SECRET_OPTIONAL_PASS ""

#define SECRET_DEVICE_KEY ""

-----#include"thingProperties.h"-----

#include <ArduinoIoTCloud.h>

#include <Arduino_ConnectionHandler.h>

const char DEVICE_LOGIN_NAME[] =
"732bb388-82ae-468e-8ee5-af94cb49f150";

const char SSID[]          = SECRET_SSID;

const char PASS[]          =
SECRET_OPTIONAL_PASS;

const char DEVICE_KEY[] =
SECRET_DEVICE_KEY;

void onPositionChange();

int Pos;

void initProperties(){

    ArduinoCloud.setBoardId(DEVICE_LOGIN_NA
    ME);

    ArduinoCloud.setSecretDeviceKey(DEVICE_KE
    Y);

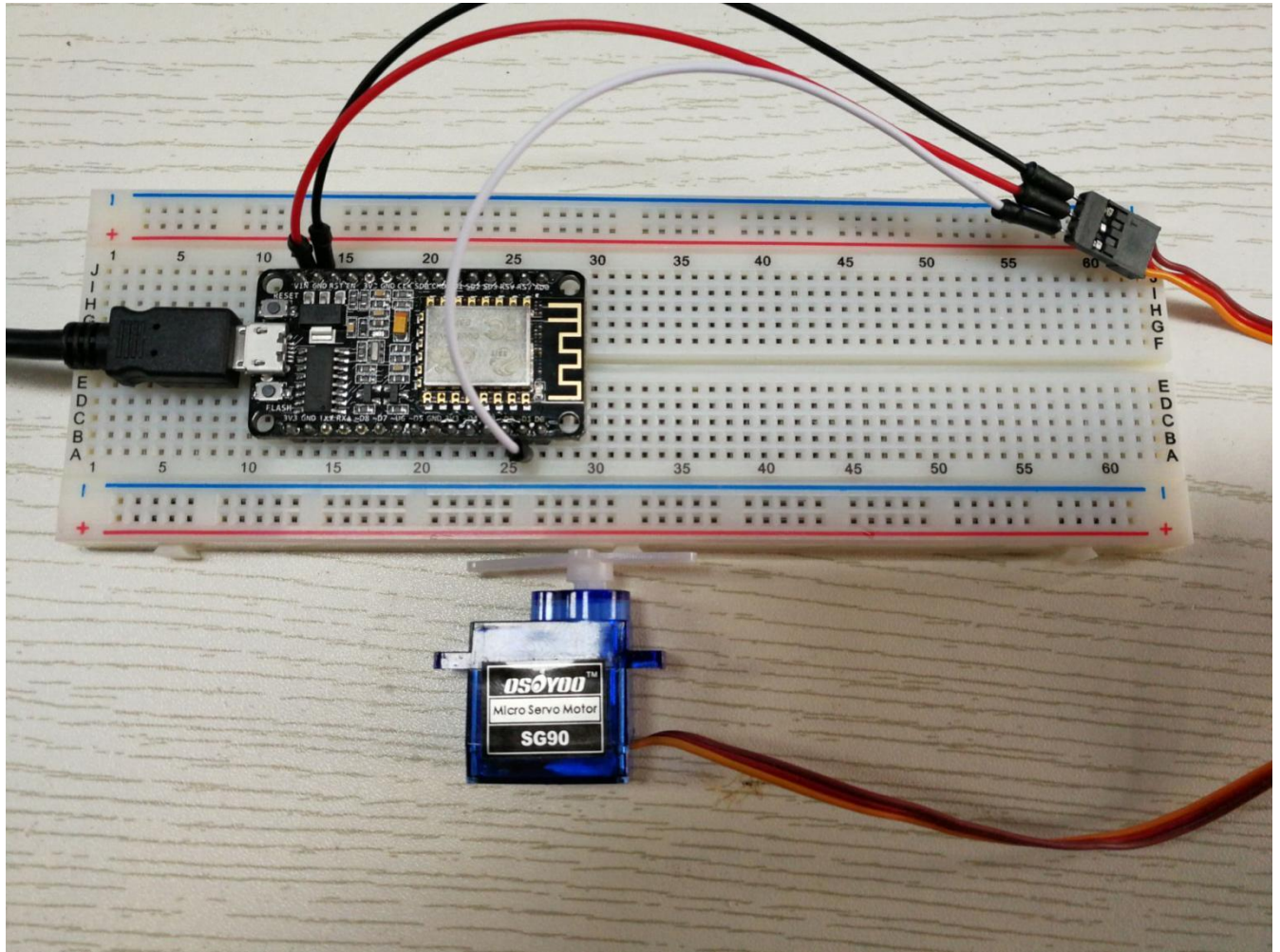
    ArduinoCloud.addProperty(Pos, READWRITE,
    ON_CHANGE, onPositionChange);

}

WiFiConnectionHandler
ArduinoIoTPreferredConnection(SSID, PASS);

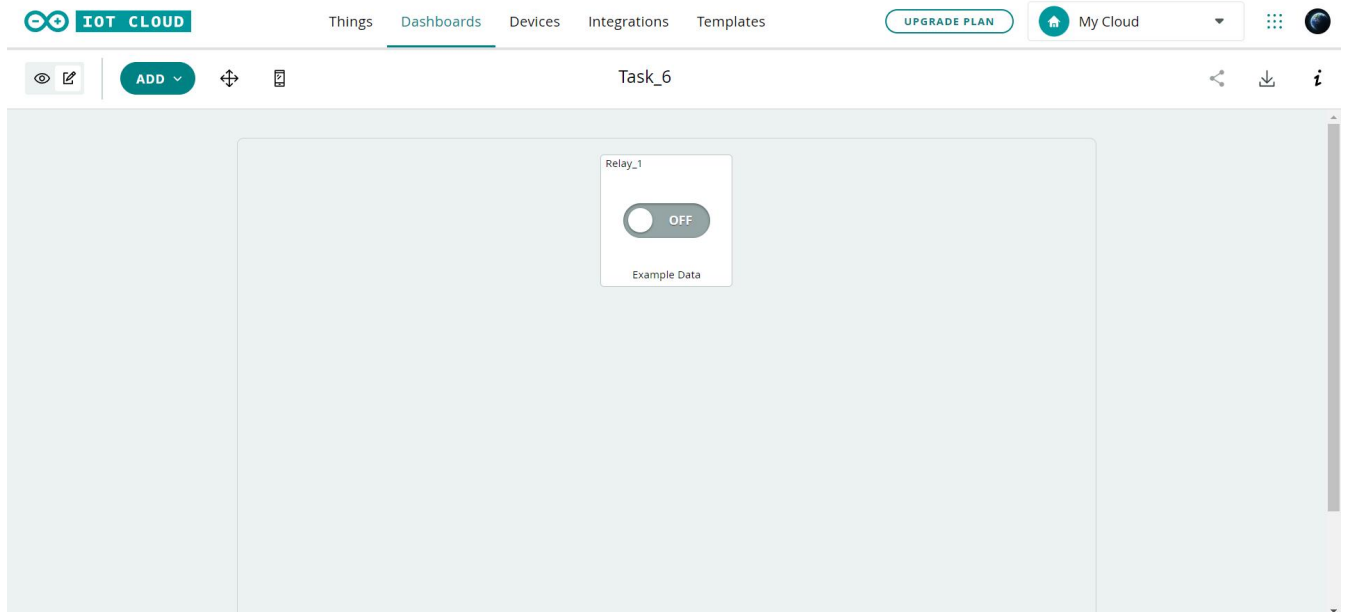
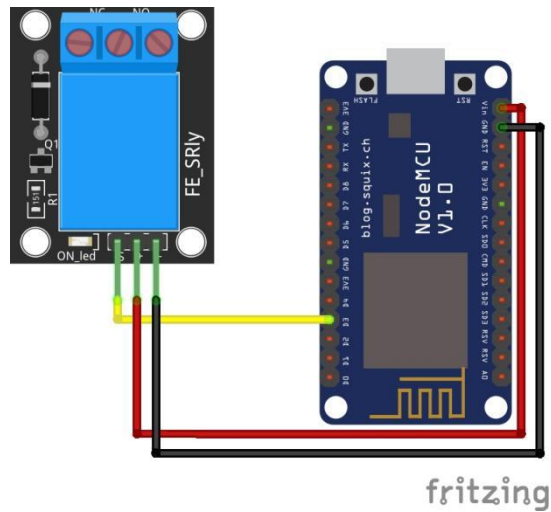
```

Output: -



4.6 Task 6: -

Control Relay Using Arduino IOT Dashboard On Android



Code:

```

#include "arduino_secrets.h"
#include "thingProperties.h"
int Rel = D3;

void setup() {
  Serial.begin(9600);
  delay(1500);
  initProperties();
  pinMode(Rel, OUTPUT);

  ArduinoCloud.begin(ArduinoIoTPreferredConnection)
;
  setDebugMessageLevel(2);
  ArduinoCloud.printDebugInfo();
}

void loop() {
  ArduinoCloud.update();
}

void onRelayChange() {
  digitalWrite(Rel, Relay);
}

-----#include"arduino_Secrets.h"-----

#define SECRET_SSID ""
#define SECRET_OPTIONAL_PASS ""
#define SECRET_DEVICE_KEY ""

-----#include"thingProperties.h"-----

#include <ArduinoIoTCloud.h>
#include <Arduino_ConnectionHandler.h>

const char DEVICE_LOGIN_NAME[] =
"732bb388-82ae-468e-8ee5-af94cb49f150";

const char SSID[] = SECRET_SSID;
const char PASS[] =
SECRET_OPTIONAL_PASS;
const char DEVICE_KEY[] =
SECRET_DEVICE_KEY;

void onLEDChange();

bool Relay;

void initProperties(){

  ArduinoCloud.setBoardId(DEVICE_LOGIN_NAME)
;
  ArduinoCloud.setSecretDeviceKey(DEVICE_KEY);
  ArduinoCloud.addProperty(Relay, READWRITE,
ON_CHANGE, onRelayChange);

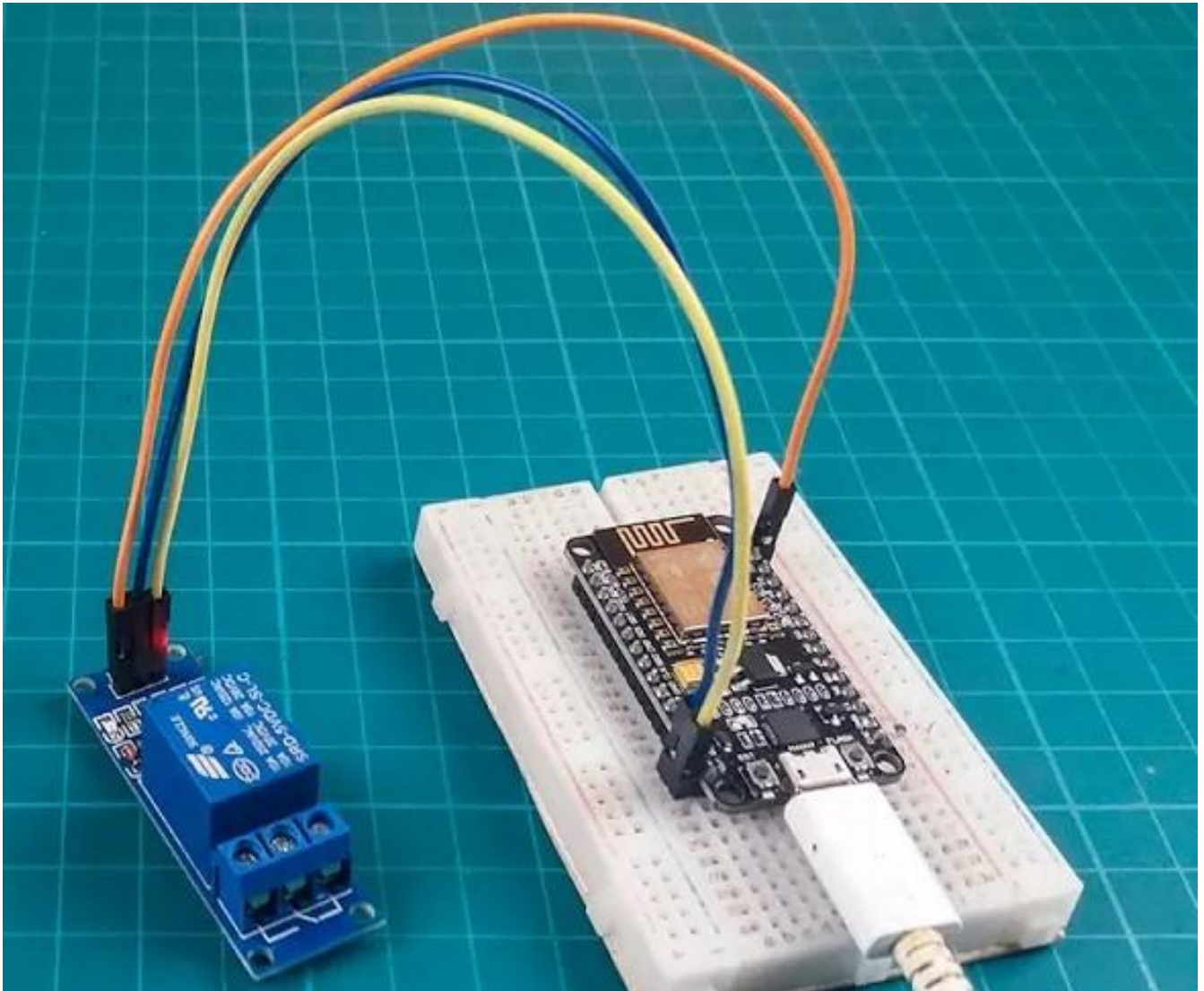
}

WiFiConnectionHandler
ArduinoIoTPreferredConnection(SSID, PASS);

```

3170001

Output: -



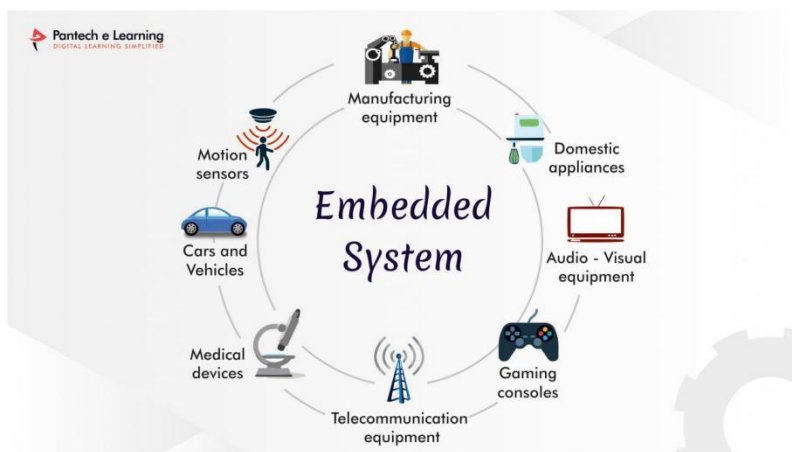
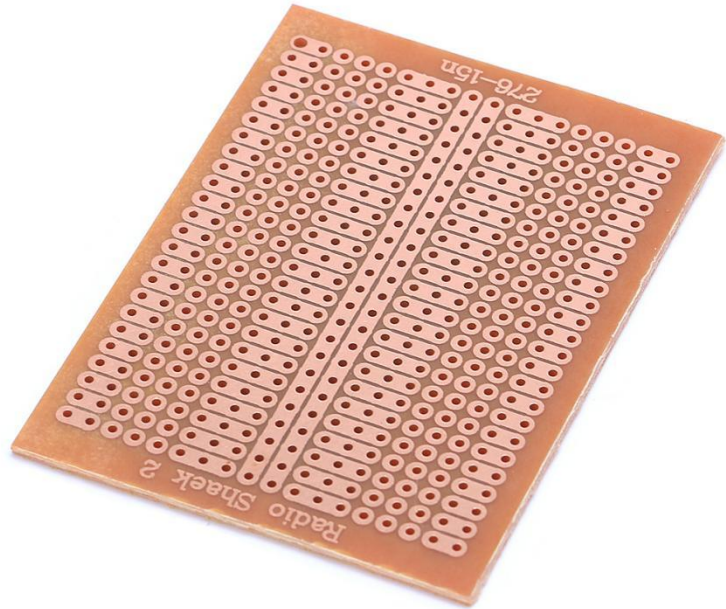
4.7 Task 7: -

Basics Of Hardware PCB, Soldering And Embedded System's

A Printed Circuit Board (PCB) is a vital component in modern electronics. It acts as a platform to interconnect electronic components, forming the foundation of various devices.

PCBs consist of thin layers of insulating material with copper traces etched onto them, creating pathways for electrical signals.

These intricate pathways enable the flow of current, allowing components like resistors, capacitors, and microchips to communicate and function together seamlessly.

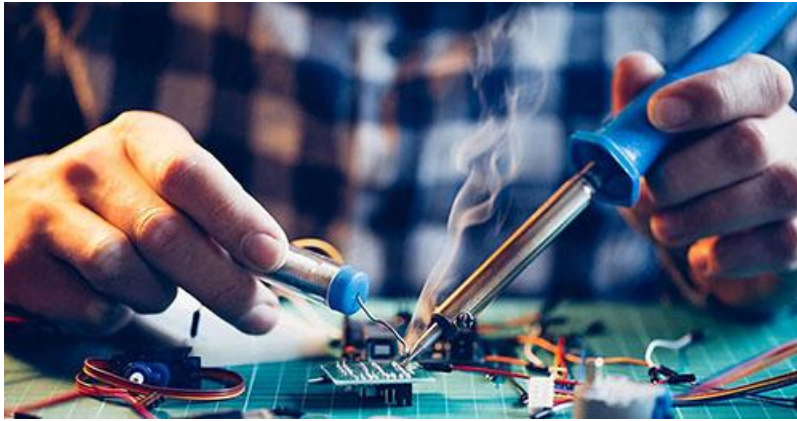


Embedded systems are integrated computing systems designed for specific tasks, seamlessly combining hardware and software for efficient and dedicated functionality.

These systems power a wide range of devices, from household appliances to industrial machinery, enhancing automation and control.

They often operate in real-time, offering reliability and responsiveness crucial for their intended applications.

The design of embedded systems involves optimizing performance, power consumption, and size to meet the unique demands of each use case.

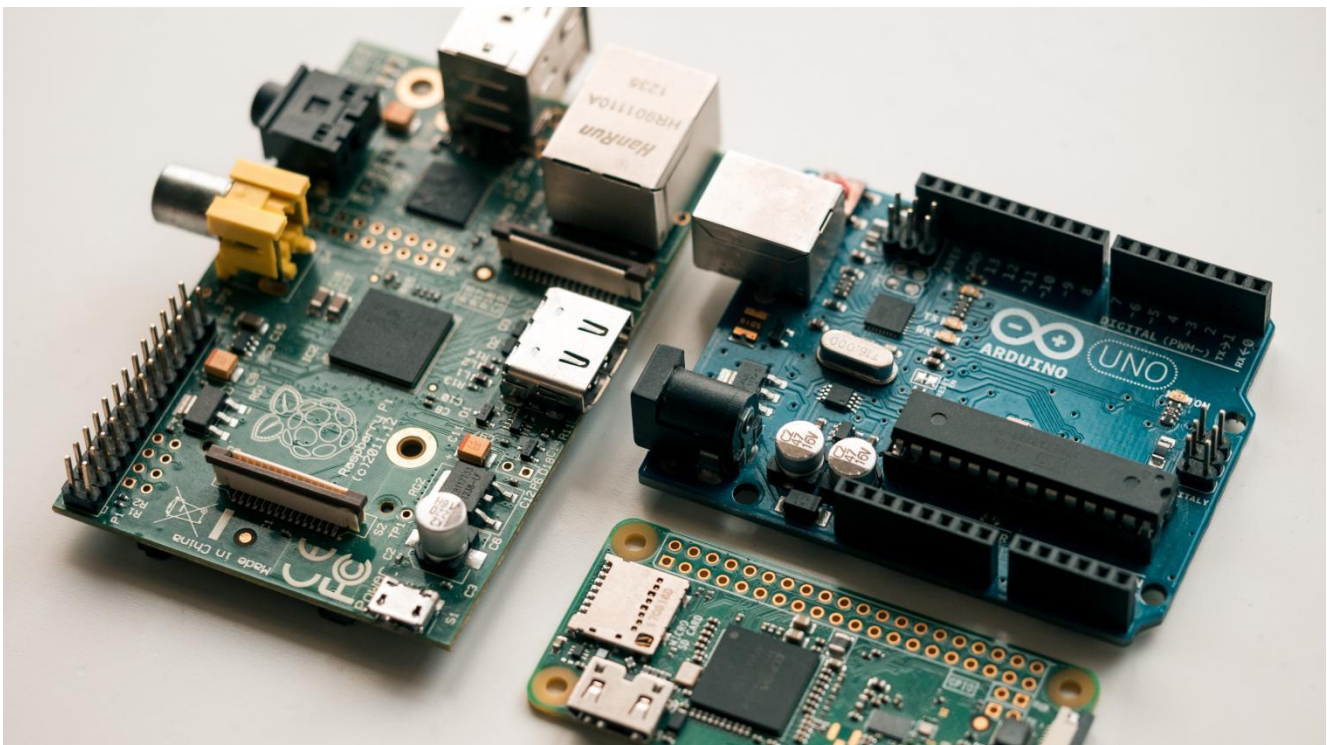


Soldering is a technique used to join electronic components by melting a filler metal (solder) onto their connection points, creating strong and reliable electrical connections essential for circuit functionality and stability.

Arduino and Raspberry Pi are both popular platforms used in the realms of cloud computing and embedded systems, albeit with different emphases.

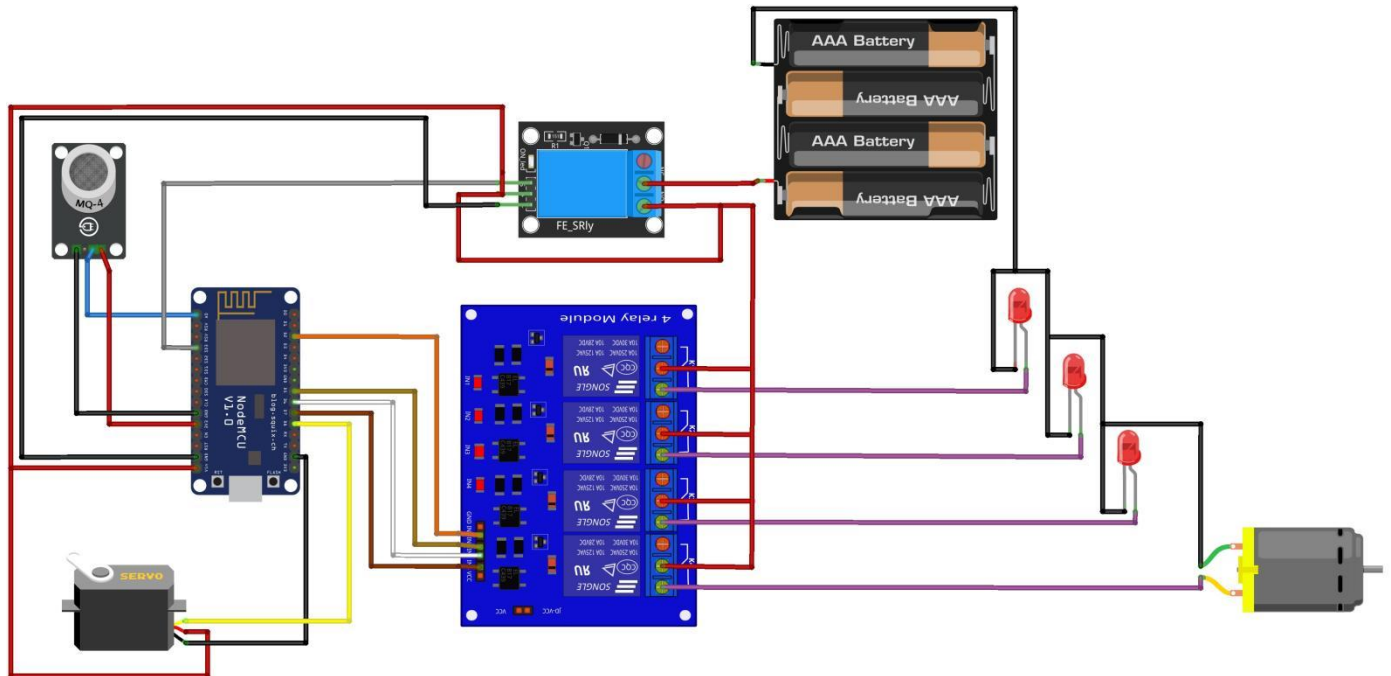
Arduino: Arduino is a microcontroller-based platform known for its simplicity and versatility in creating embedded systems. It's well-suited for projects where real-time interaction and control are critical. While Arduino itself doesn't have built-in networking capabilities for cloud integration, you can extend its functionality by adding shields or modules to enable Wi-Fi or Ethernet connectivity. This allows Arduino-based projects to interact with cloud services, sending and receiving data. Arduino is often chosen for projects that require dedicated control and responsiveness, such as sensor monitoring, robotics, and automation.

Raspberry Pi: Raspberry Pi, on the other hand, is a single-board computer that offers more processing power, memory, and connectivity options compared to Arduino. Raspberry Pi is highly capable of running full-fledged operating systems and applications, making it suitable for cloud-related tasks. It can run services that interact with cloud platforms like AWS, Azure, or Google Cloud. Raspberry Pi's stronger computational abilities make it a great choice for tasks like data processing, web hosting, and IoT gateways, where cloud connectivity and data analysis are prominent.



4.8 Final Task:

Build a Personal Home Automation Project Using Arduino Cloud



Code:

Home_Automation_aug09a.ino

```
#include "arduino_secrets.h"
#include "thingProperties.h"
```

```
Servo myservo;// create servo object to control a servo
// twelve servo objects can be created on most boards
```

```
// define the GPIO connected with Relays
#define RelayPin1 13 //D1
#define RelayPin2 4 //D2
#define RelayPin3 14 //D5
#define RelayPin4 12 //D6
#define Safety 10 //SD3
```

```

#define wifiLed 16 //D0

const int sensorPin = A0;

void setup() {
  // Initialize serial and wait for port to open:
  Serial.begin(9600);
  // This delay gives the chance to wait for a Serial Monitor without blocking
  if none is found
  delay(1500);

  // Defined in thingProperties.h
  initProperties();

  // Connect to Arduino IoT Cloud
  ArduinoCloud.begin(ArduinoIoTPreferredConnection);

  setDebugMessageLevel(2);
  ArduinoCloud.printDebugInfo();

  myservo.attach(15); // attaches the servo on GIO2 to the servo object

  pinMode(RelayPin1, OUTPUT);
  pinMode(RelayPin2, OUTPUT);
  pinMode(RelayPin3, OUTPUT);
  pinMode(RelayPin4, OUTPUT);
  pinMode(Safety, OUTPUT);

  pinMode(wifiLed, OUTPUT);

  // Set up the sensor pin as an input
  pinMode(sensorPin, INPUT);

  //During Starting all Relays should TURN OFF
  digitalWrite(RelayPin1, HIGH);
  digitalWrite(RelayPin2, HIGH);

```

```
digitalWrite(RelayPin3, HIGH);
digitalWrite(RelayPin4, HIGH);

digitalWrite(wifiLed, HIGH); //Turn OFF WiFi LED
}

void loop() {
  ArduinoCloud.update();

  //  manual_control(); Control relays manually

  int sensorValue = analogRead(sensorPin);
  Serial.print("Sensor Value: ");
  Serial.println(sensorValue);
  gAS = sensorValue;

  //Sensor Safety
  if (sensorValue > 75) {
    switch_1 = false;
    switch_2 = false;
    switch_3 = false;
    switch_4 = false;
    digitalWrite(RelayPin1, LOW);
    digitalWrite(RelayPin2, LOW);
    digitalWrite(RelayPin3, LOW);
    digitalWrite(RelayPin4, LOW);
    myservo.write(180);
    delay(2000);
    digitalWrite(Safety, LOW);
  }
  else{
    digitalWrite(Safety, HIGH);
    myservo.write(0);
  }

  if (WiFi.status() != WL_CONNECTED)
```

```
{  
  digitalWrite(wifiLed, HIGH); //Turn OFF WiFi LED  
}  
else{  
  digitalWrite(wifiLed, LOW); //Turn ON WiFi LED  
}  
}
```

```
void onSwitch1Change() {  
  if (switch_1 == 1)  
  {  
    digitalWrite(RelayPin1, HIGH);  
    Serial.println("Device1 ON");  
  }  
  else  
  {  
    digitalWrite(RelayPin1, LOW);  
    Serial.println("Device1 OFF");  
  }  
}
```

```
void onSwitch2Change() {  
  if (switch_2 == 1)  
  {  
    digitalWrite(RelayPin2, HIGH);  
    Serial.println("Device2 ON");  
  }  
  else  
  {  
    digitalWrite(RelayPin2, LOW);  
    Serial.println("Device2 OFF");  
  }  
}
```

```
void onSwitch3Change() {  
  if (switch_3 == 1)  
  {  
    digitalWrite(RelayPin3, HIGH);  
    Serial.println("Device2 ON");  
  }  
  else  
  {  
    digitalWrite(RelayPin3, LOW);  
    Serial.println("Device3 OFF");  
  }  
}
```

```
void onSwitch4Change() {  
  if (switch_4 == 1)  
  {  
    digitalWrite(RelayPin4, HIGH);  
    Serial.println("Device4 ON");  
  }  
  else  
  {  
    digitalWrite(RelayPin4, LOW);  
    Serial.println("Device4 OFF");  
  }  
}
```

```
void onGASChange() {  
  // Add your code here to act upon GAS change  
}
```

Arduino_secrets.h

```
#define SECRET_SSID "Your Wifi Credentials"
#define SECRET_OPTIONAL_PASS "Your WiFi Password"
#define SECRET_DEVICE_KEY "Device Key Provided During Cloud Linking"
```

ThingProperties.h

// Code generated by Arduino IoT Cloud, DO NOT EDIT.

```
#include <ArduinoIoTCloud.h>
#include <Arduino_ConnectionHandler.h>
#include <Servo.h>

//const char THING_ID[] = "THING ID"; //Enter THING ID
const char DEVICE_LOGIN_NAME[] = "4c3254c6-b599-49e1-89ba-67d3ed12441a";

const char SSID[] = SECRET_SSID; // Network SSID (name)
const char PASS[] = SECRET_OPTIONAL_PASS; // Network password (use
for WPA, or use as key for WEP)
const char DEVICE_KEY[] = SECRET_DEVICE_KEY; // Secret device password

void onSwitch1Change();
void onSwitch2Change();
void onSwitch3Change();
void onSwitch4Change();
void onGASChange();

CloudSwitch switch_1;
CloudSwitch switch_2;
CloudSwitch switch_3;
CloudSwitch switch_4;
int gAS;

void initProperties(){

  ArduinoCloud.setBoardId(DEVICE_LOGIN_NAME);
  ArduinoCloud.setSecretDeviceKey(DEVICE_KEY);
  ArduinoCloud.addProperty(switch_1, READWRITE, ON_CHANGE,
```

```

onSwitch1Change);
  ArduinoCloud.addProperty(switch_2, READWRITE, ON_CHANGE,
onSwitch2Change);
  ArduinoCloud.addProperty(switch_3, READWRITE, ON_CHANGE,
onSwitch3Change);
  ArduinoCloud.addProperty(switch_4, READWRITE, ON_CHANGE,
onSwitch4Change);
  ArduinoCloud.addProperty(gAS, READWRITE, ON_CHANGE, onGASChange);

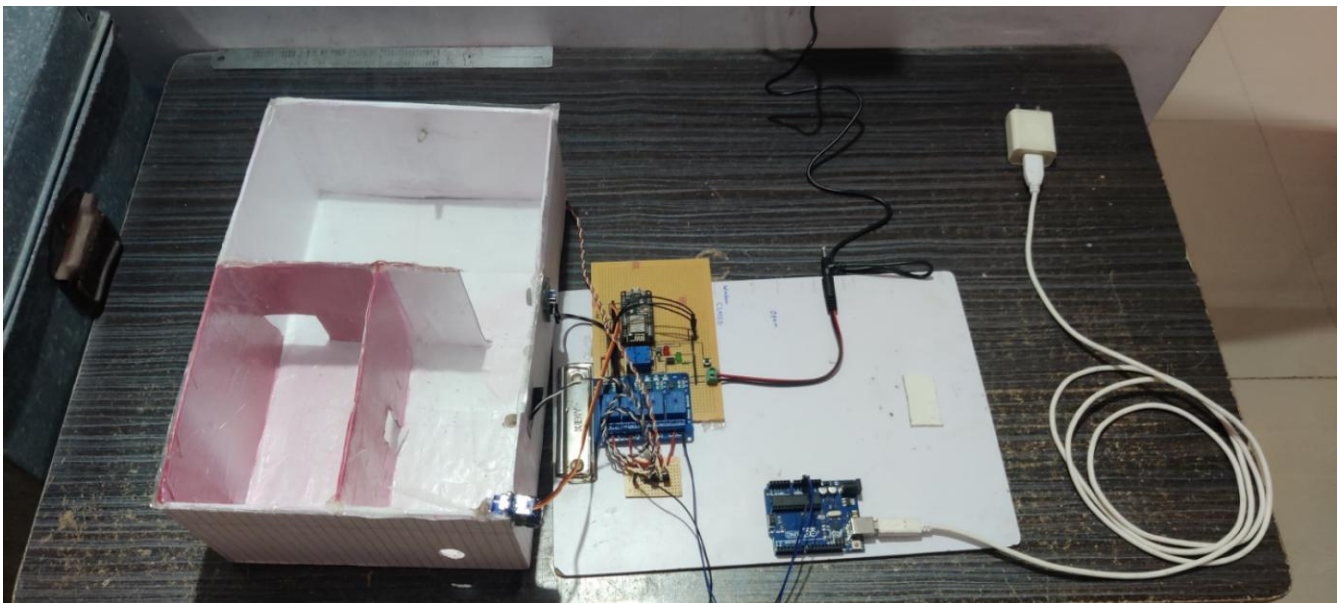
}

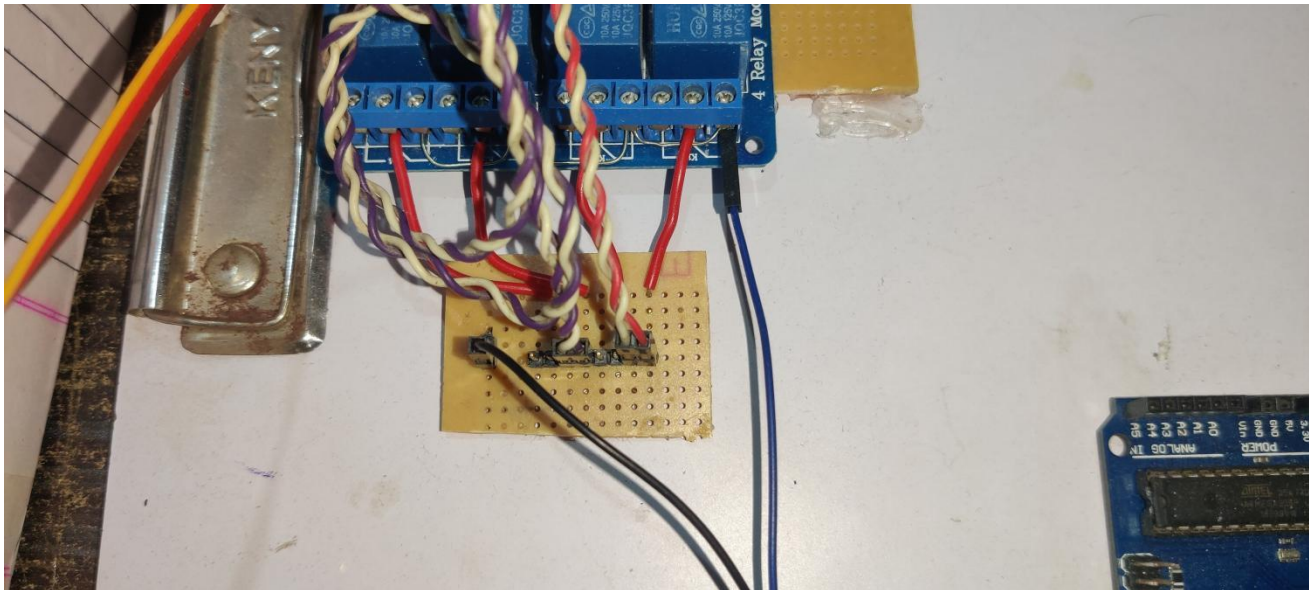
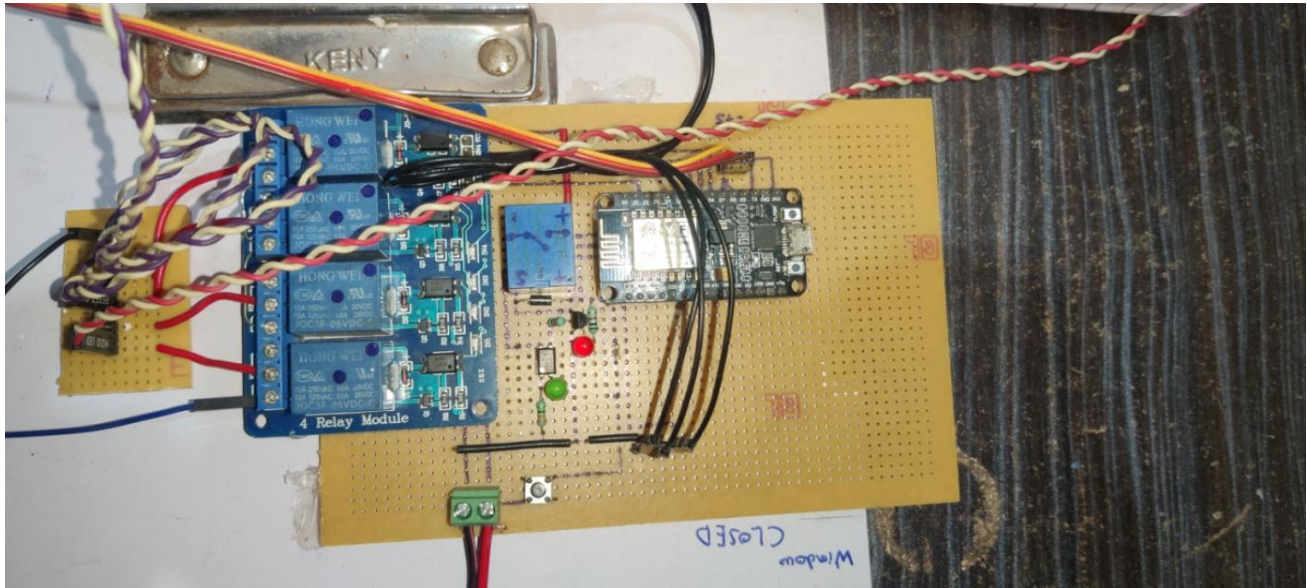
```

```
WiFiConnectionHandler ArduinoIoTPreferredConnection(SSID, PASS);
```

Output:

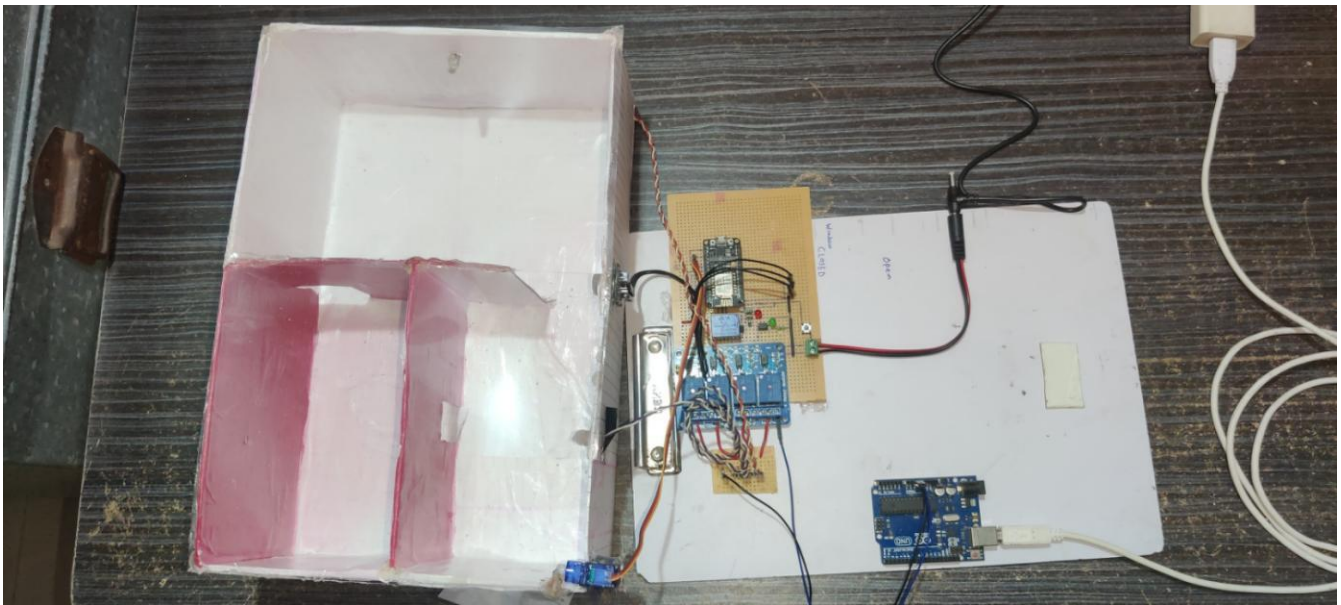
Home Automation using Arduino Cloud and Alexa represents a seamless convergence of cutting-edge technologies for modern living. This synergy enables users to effortlessly manage various smart devices—such as lights, thermostats, and appliances—using simple voice commands through Alexa. The integration also allows for remote monitoring and control. This innovative combination of Arduino Cloud and Alexa showcases the power of IoT which provides a glimpse into the future of interconnected smart homes, where technology works harmoniously to simplify and elevate daily life.







The Smoke Sensor Detects the LPG level and the Smoke in The House in Case of fire in the house. The Control mechanism will take the safety measures which will open the Window and turn off the Whole lighting in the House.



Output Live Video :- //Yet To Shoot, Will be on YouTube.

CONCLUSION

In conclusion with acceptance that the whole project could not be done without our internal guide as well as the guidance of the industry Professional's.

In conclusion, my internship experience in the field of Internet of Things (IoT) and Arduino Cloud, along with the completion of a Home Automation Mini Project, has been incredibly rewarding and enlightening. Throughout the internship, I had the opportunity to delve into the exciting realm of IoT technology and its practical applications, particularly in the context of home automation.

Over the course of the internship, I acquired a deep understanding of IoT principles, protocols, and the seamless integration of hardware and software components. Through hands-on experience with Arduino Cloud, I gained valuable insights into remote monitoring, data analysis, and control mechanisms, which are vital aspects of modern IoT systems. This exposure has not only expanded my technical skill set but also heightened my awareness of the immense potential IoT holds for transforming industries and enhancing everyday life.

I am sincerely grateful for the guidance and mentorship provided by the Internal Guide and experts in the company throughout this internship. Their insights, feedback, and support were invaluable in shaping my understanding and enhancing my skills. This experience has left me with a sense of accomplishment and enthusiasm, and I am eager to embark on new challenges and contribute to the exciting advancements in IoT technology.

5. REFERENCES LINKS

Arduino Cloud Documentation : <https://docs.arduino.cc/arduino-cloud/>

IOT Dashboard: <https://create.arduino.cc/iot/dashboards>

YouTube: <http://www.youtube.com>